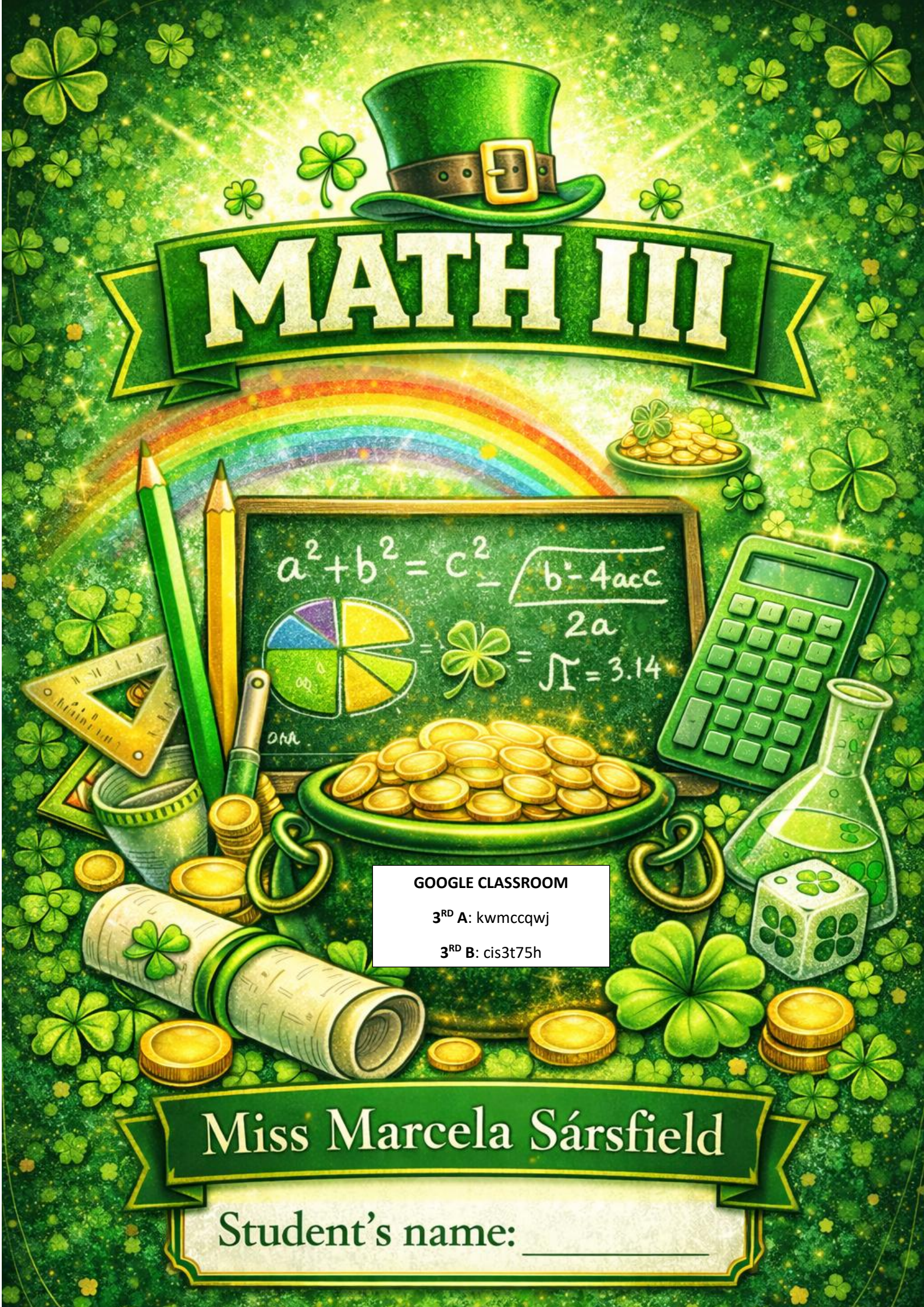

MATH III
$a^2+b^2=c^2$
$\frac{-b \pm \sqrt{b^2-4ac}}{2a}$
$\pi = 3.14$
GOOGLE CLASSROOM
3RD A: kwmccqwj
3RD B: cis3t75h
Miss Marcela Sársfield
Student's name: _____

Subject materials:



- ⊕ **Folder with squared sheets.**
- ⊕ **Pencil, eraser, protactor, and ruler.**
- ⊕ **Calculator**
- ⊕ **Printed booklet.**

EVERYONE SHOULD HAVE THE MATERIALS FOR ALL CLASSES.

TESTS

Date	Topics

Course: 3rd year

Teachers: Marcela Sársfield Arias

Course Objective

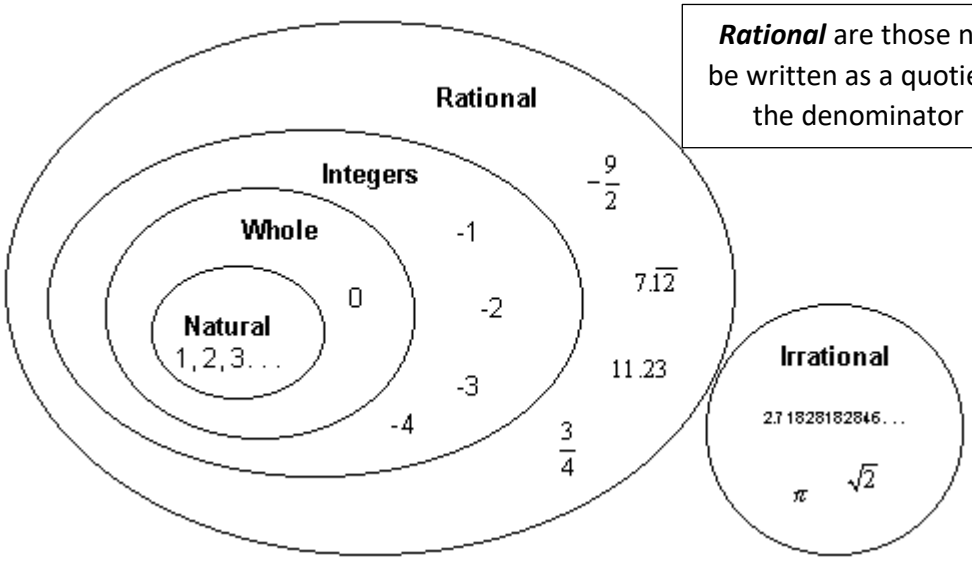
- Get the students prepared to take the S.A.T.
- Understand the significance and meaningful concepts, as well as connections among numeric fields or geometric concepts.
- Be able to transfer to daily life the contents given (including topics, problems and issues related to cross-curricular projects: Sustainable Development Goals and E.S.I.)
- Integrate the knowledge acquired in previous years with the new information.
- Promote respect to diversity and equality (E.S.I)

Unit Titles/Topics/Brief Description

Unit 0	**DIAGNOSTIC**
• Rational numbers. • Fractions. Proper, improper, mixed numbers. Operations with fractions. • Operations with rational numbers. Properties of operations. Order of operations. (PEMDAS). • Roots and exponents. Rules. • Arithmetic word problems	
Unit 1	**NUMBERS AND OPERATIONS**
• Rational numbers. Decimals. • Place Value. • Scientific notation • Ratios, proportions, and percent	
Unit 2	**GEOMETRY AND MEASUREMENT**
• Triangles. Equilateral and isosceles triangles. Right triangles and Pythagorean Theorem.	
Unit 3	**ALGEBRA AND FUNCTIONS**
• Relations/Functions. • Curve sketching • Linear equations with one variable. Linear graphs. Difference between equation and function.	
Unit 4	**DATA ANALYSIS, PROBABILITY AND STADISTICS**
• Mean, media, and mode. • Interpretation of information presented in stadistical graphs. • Combinations and Permutations	

DIAGNOSTIC UNIT

RATIONAL NUMBERS VS REAL NUMBERS



Rational are those numbers which can be written as a quotient of two integers, the denominator being non-zero.

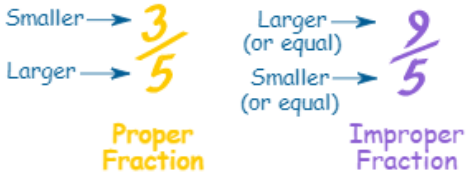
FRACTIONS

Numerator / Denominator

We call the top number the **Numerator**, it is the number of parts we **have**.
We call the bottom number the **Denominator**, it is the number of parts the whole is **divided into**.

$$\frac{\text{Numerator}}{\text{Denominator}}$$

Types of Fractions



Proper Fractions:	The numerator is less than the denominator
	Examples: $^1/_3$, $^3/_4$, $^2/_7$

Improper Fractions:	The numerator is greater than (or equal to) the denominator
	Examples: $^4/_3$, $^{11}/_4$, $^7/_7$

→ *OPERATIONS WITH FRACTIONS*

ADDITION/SUBTRACTION

To add fractions there are Three Simple Steps:

- Step 1: Make sure the bottom numbers (the **denominators**) are the same
- Step 2: Add the top numbers (the **numerators**), put that answer over the denominator
- Step 3: Simplify the fraction (if possible)

There are 3 simple steps to subtract fractions

- Step 1. Make sure the bottom numbers (the denominators) are the same
- Step 2. Subtract the top numbers (the numerators). Put the answer over the same denominator.
- Step 3. Simplify the fraction (if needed).

MULTIPLICATION

1. Multiply the top numbers (the *numerators*).
2. Multiply the bottom numbers (the *denominators*).
3. Simplify the fraction if needed.

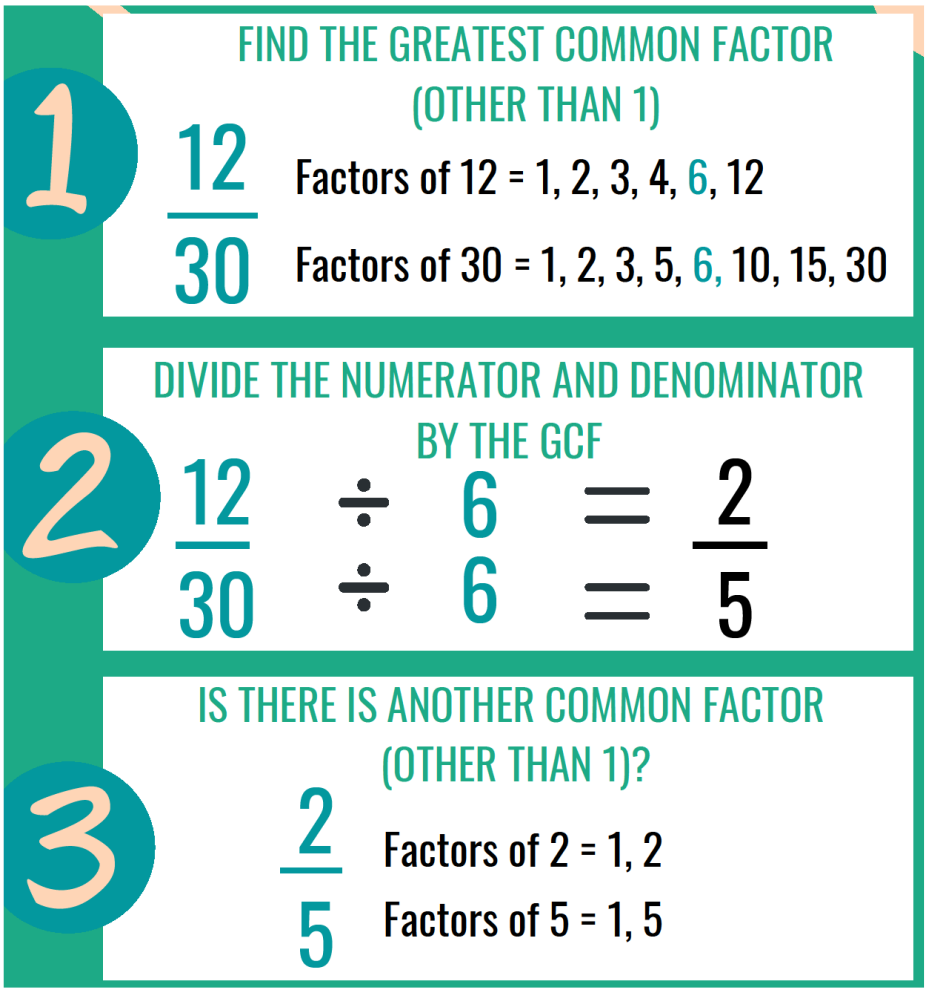
DIVISION

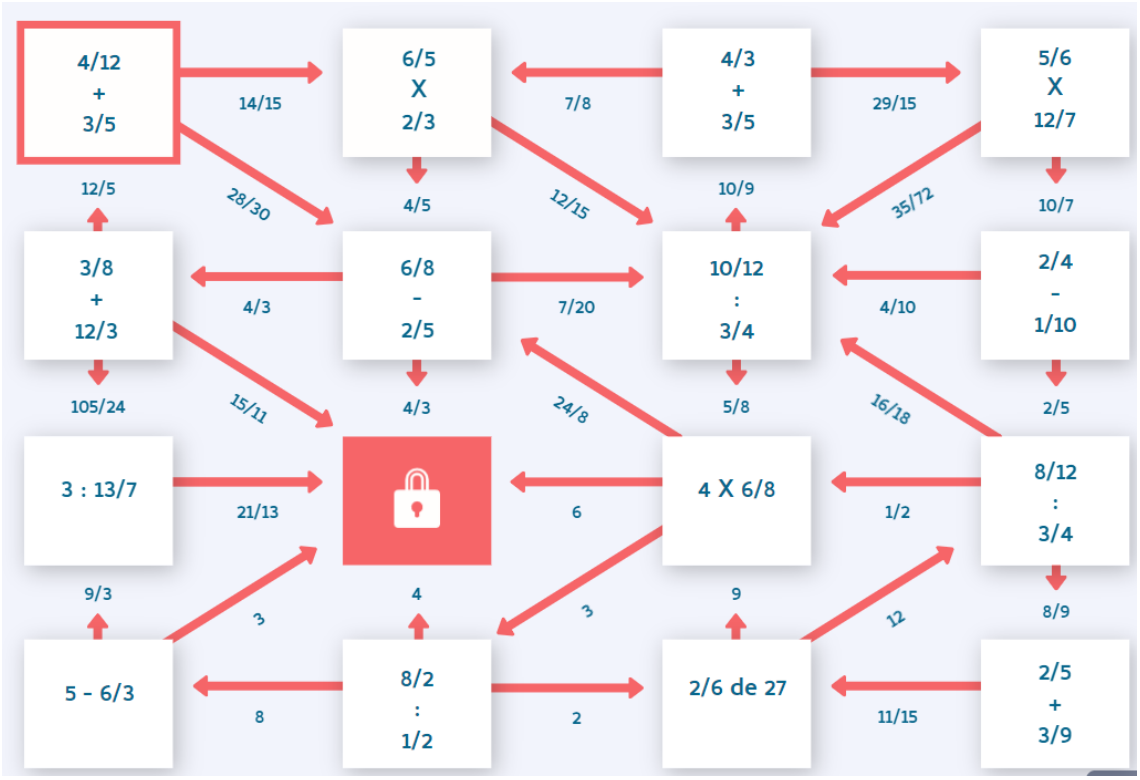
Step 1. Turn the second fraction *(the one you want to divide by)* upside down (this is now a reciprocal).

Step 2. Multiply the first fraction by that reciprocal

Step 3. Simplify the fraction (if needed)

HOW TO SIMPLIFY A FRACTION





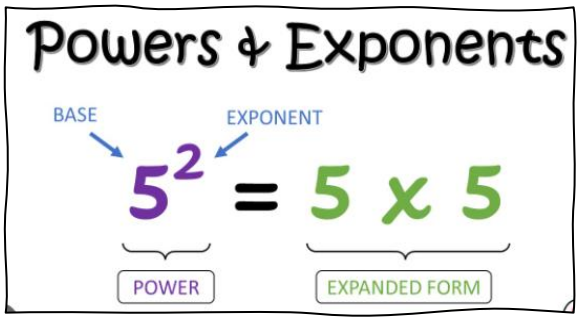
WORD PROBLEMS

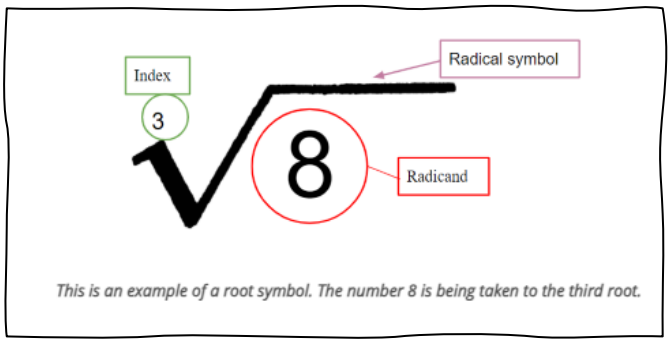
1. One half of the students in a school are girls, 3/5 of these girls are studying in lower classes. What fraction of students are girls and are studying in lower classes?
2. According to the previous exercise, what fraction of girls are studying in upper classes?
3. Maddy reads three-fifth of 75 pages of his lesson. How many more pages does he need to read to complete the lesson?
4. A herd of cows gives 4 liters of milk each day. But each cow gives one-third of total milk each day. They give 24 liters milk in six days, how many cows are there in the herd?
5. Addie bought 9 candy bars and ate 1/3 of them. Frannie bought 6 candy bars and ate 2/3 of them. Who ate more candy bars?
   a) Addie
   b) Frannie
   c) They ate the same number of candy bars

6. Raul bought 6 tangerines and ate 2/3 of them. Omar bought 8 tangerines and ate 1/4 of them. Who ate more tangerines?
   a) Raul
   b) Omar
   c) They ate the same number of tangerines

7. If 24 trousers of equal size can be prepared in 54 meters of cloth, what length of the cloth is required for each trouser?

ROOTS AND EXPONENTS





Law	Example
$a^m a^n = a^{m+n}$	$2^3 2^4 = 2^{3+4} = 2^7 = 128$
$(a^m)^n = a^{mn}$	$(2^3)^4 = 2^{3.4} = 2^{12} = 4096$
$(ab)^n = a^n b^n$	$(20)^3 = (2.10)^3 = 2^3.10^3 = 8.1000 = 8000$
$(\frac{a}{b})^n = \frac{a^n}{b^n}$	$(\frac{2}{5})^3 = \frac{2^3}{5^3} = \frac{8}{125}$
$\frac{a^m}{a^n} = a^{m-n}$	$\frac{2^5}{2^3} = 2^{5-3} = 2^2 = 4$
$\frac{a^m}{a^n} = \frac{1}{a^{n-m}}$	$\frac{2^3}{2^5} = \frac{1}{2^{5-3}} = \frac{1}{2^2} = \frac{1}{4}$

EXPONENT PROPERTIES - MIXED REVIEW SCAVENGER HUNT WORKSHEET

Name: _____

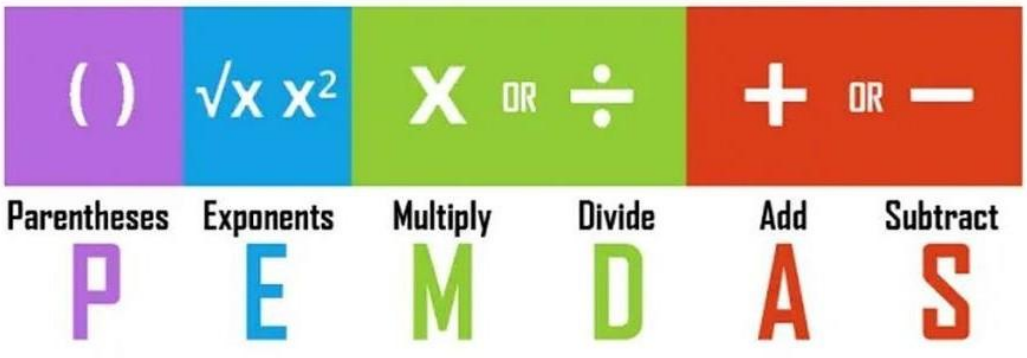
Hour: _____

Solve the first problem in the box below where it says START. The answer to this problem will lead you to the next problem to be completed. Your answer should be in one of the gray rectangular boxes. Complete the next problem (place a 2 in the starburst bubble) and so on until you have finished all twelve problems. Your last problem should be the answer in the start box.

START y^7 (Your last answer)	**IF YOU FOUND ...** $(x+4)^6$	**IF YOU FOUND ...** x^6
$(x^4)^2$ 1	$x^3 \bullet x^? = x^{17}$	$(x\frac{1}{2}y)^4$
IF YOU FOUND ... $\frac{1}{9x^2}$	**IF YOU FOUND ...** $108x^{16}$	**IF YOU FOUND ...** $\frac{y^3}{x^3}$
$(\frac{4x^2}{x^3})^? = \frac{16}{x^2}$	$7x^0$	$\frac{1}{3^2} \bullet 3^5$
IF YOU FOUND ... x^2y^4	**IF YOU FOUND ...** 2	**IF YOU FOUND ...** 27
$(2x^5)^2(3x^2)^3$	$(\frac{x}{y})^{-3}$	$y^2 \bullet y^5 \bullet y^0$
IF YOU FOUND ... 7	**IF YOU FOUND ...** 14	**IF YOU FOUND ...** x^8
$[(x+4)^2]^3$	$(-3x)^{-2}$	$\frac{x^8}{x^2}$

ORDER OF OPERATIONS

The following convention has been commonly adopted, when evaluating arithmeticexpressions, the order of operations is:



Remember: If a numerical expression includes a fraction bar, perform all calculations above and below the fraction bar before dividing the top by the bottom number.

EXERCISES:



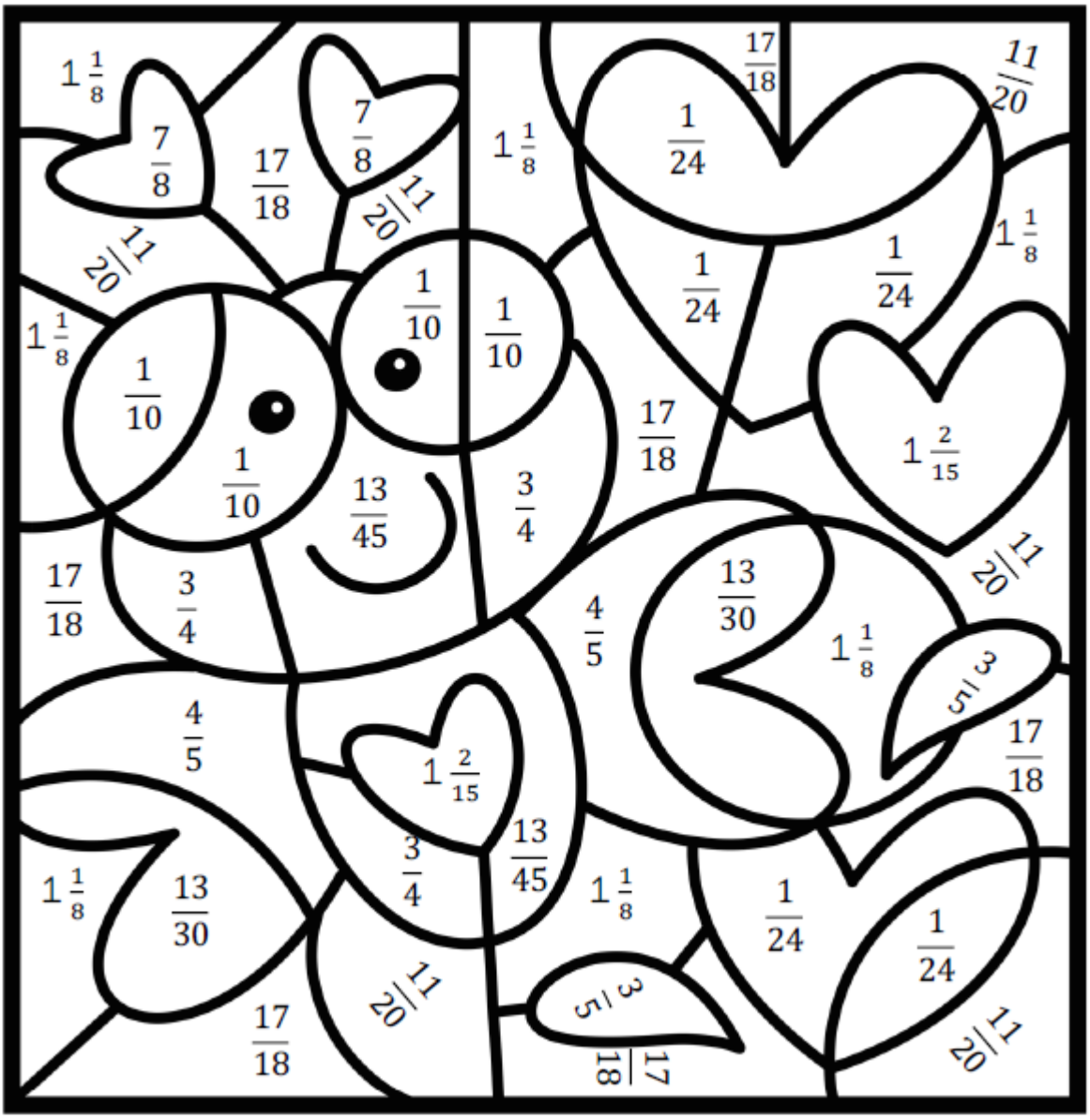
Name _____Date_____

ORDER OF OPERATIONS With Fractions



Directions: Evaluate each expression using the order of operations. Color the answer on the color sheet the color at the bottom of each box.

1) $\frac{3}{5} \times (\frac{1}{4} + \frac{3}{4}) =$ Color orange.	2) $\frac{7}{9} + \frac{4}{5} \times \frac{4}{9} =$ Color yellow.	3) $\frac{1}{8} \div \frac{1}{5} + \frac{1}{2} =$ Color purple.	4) $\frac{1}{4} \times \frac{5}{6} - \frac{1}{6} =$ Color green.
5) $\frac{4}{5} \times \frac{1}{4} + \frac{3}{5} =$ Color blue.	6) $(\frac{2}{9} + \frac{1}{2}) \times \frac{2}{5} =$ Color pink.	7) $\frac{7}{8} \times (\frac{1}{6} + \frac{5}{6}) =$ Color red.	8) $\frac{1}{9} \div \frac{1}{4} + \frac{1}{2} =$ Color purple.
9) $\frac{3}{5} - \frac{1}{3} \times \frac{1}{2} =$ Color blue.	10) $\frac{1}{2} + \frac{1}{3} \times \frac{3}{4} =$ Color pink.	11) $\frac{3}{5} \times \frac{1}{2} - \frac{1}{5} =$ Color white.	12) $\frac{2}{5} + \frac{1}{4} \times \frac{3}{5} =$ Color purple.



1. Solve each expression using the correct order of operations.

a) $((-5)-5)\cdot\frac{10}{-2}$

b) $5\cdot 6-1-\frac{6}{(-2)\cdot(-1)}$

c) $\left(\frac{1}{9}\times\frac{3}{5}\right)\div\left(\frac{7}{9}+\frac{1}{8}-\frac{3}{4}\right)\times\left(\frac{3}{8}+\frac{1}{6}\right)$

d) $\left(\left(\frac{1}{4}+\frac{1}{2}\right)\div\frac{4}{5}\right)\times\frac{8}{9}-\frac{5}{6}+\frac{1}{5}\times\frac{3}{8}$

e) $\frac{1}{6}+\frac{8}{9}\times\left(\frac{3}{8}-\frac{1}{8}\right)\div\frac{1}{4}\div\left(\frac{4}{5}+\frac{1}{5}\right)$

f) $\frac{5}{8}\times\frac{4}{5}-\frac{2}{9}+\frac{3}{4}\div\left(\frac{2}{5}\div\left(\frac{7}{9}\times\frac{3}{5}\right)\right)$

g) $\left(\sqrt{\frac{1}{9}}-\left[\left(\frac{1}{3}\right)^2:\left(\frac{1}{2}\right)^2\times 1.5-\left(0.5+2\frac{2}{3}\right)+0.5\right]^3+\frac{2}{3}\right)$

2. Rewrite the words as a numerical expression.

a) Find de number that is 6 times the sum of 4 and 9.

b) Subtract 7 from the product of 8 and 6.

c) Divide the difference between 250 and 200 by 10.

d) Double the sum of 15 and 8, then subtract 20.

e) Multiply the sum of 7 and 6 by the sum of 4 and 5.

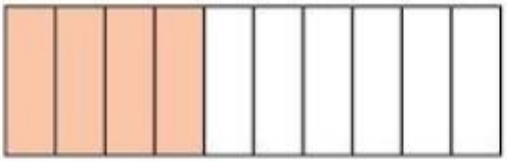
PUZZLE

$(-2) \cdot (+9) - (-4) =$	$\frac{3}{8} : \frac{5}{7} =$	$\frac{3}{5} - \frac{3}{7} =$
$3 \cdot \frac{-2}{9} =$	$\frac{4}{3} + 6 =$	$(-2) \cdot 5 - 4 \cdot \sqrt{25} + 2^3 =$
$35 - (16 + 9) - 3 =$	$3 \cdot 4 + 12 : 6 =$	$\frac{3}{6} + \frac{5}{6} + \frac{8}{6} =$
$\frac{9}{6} + \frac{8}{12} - \frac{7}{8} =$	$\sqrt{64^2} - (6^2 : 3) - \frac{8}{2} =$	$\frac{5}{6} \cdot \frac{8}{9} =$

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UNIT ONE: NUMBERS AND OPERATIONS

DECIMALS



$\frac{4}{10}$ of the rectangle is shaded.

Tenths can be written a shorter way:

$$\frac{4}{10} = 0.4$$

no units — 0 — decimal point — . — 4 — 4 tenths

0.4 is the **decimal form** for the fraction $\frac{4}{10}$.

The · is called the **decimal point**.

It separates the whole numbers from the tenths.

EXAMPLE 4

Write as decimals

a $2\frac{3}{10}$ **b** $6\frac{7}{10}$

a $2\frac{3}{10} = 2.3$

b $6\frac{7}{10} = 6.7$

Exercise 1D

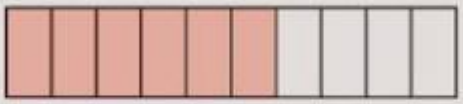
1 Write these fractions in decimal form.

a $\frac{3}{10}$ **b** $\frac{6}{10}$ **c** $\frac{9}{10}$ **d** $1\frac{2}{10}$

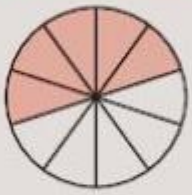
e $2\frac{8}{10}$ **f** $3\frac{4}{10}$ **g** $12\frac{1}{10}$ **h** $132\frac{5}{10}$

2 Write the part of each shape that is shaded – first as a fraction, then as a decimal:

a



b



3 Write in order of size, smallest first.

a 0.3, 0.6, 0.2, 0.8

b 0.9, 0.5, 1, 0

c 2.1, 0.6, 0.9, 1.8

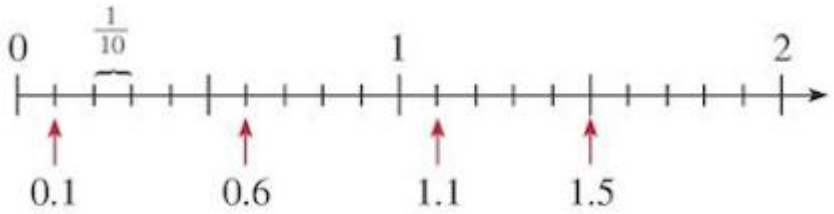
4 What number is $\frac{1}{10}$ more than

a 0.3 **b** 0.7 **c** 0.9

d 1 **e** 2.9 **f** 6?

Representing decimals on a number line

This number line has been divided into tenths.



The distance between each division is *one* tenth.

Exercise 1E

1 Copy the number line:



Show these numbers on your number line.

a 0.3 **b** 1.3 **c** 1.9

2 Write as decimals the letters marked on the number line.



3 What is the temperature, in °C, on these thermometers?

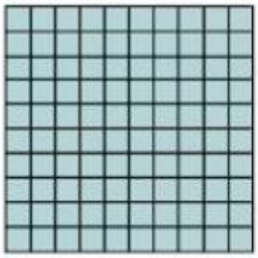
a



b



Two decimal places

If  represents 1 unit

then  represents $\frac{1}{10}$ (one tenth)

and  represents $\frac{1}{100}$ (one hundredth)

Hundredths can be written a shorter way as decimals:

$\frac{1}{100} = 0.01$

0 : no units
. : decimal point
0 : no tenths
1 : 1 hundredth

Exercise 1F

1 Use diagrams to show these decimals.

a	0.2	**b**	0.07	**c**	0.12
d	0.36	**e**	1.2	**f**	1.23

2 What decimals do these pictures represent?

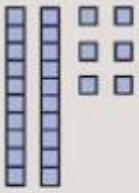
a



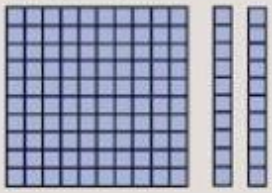
b



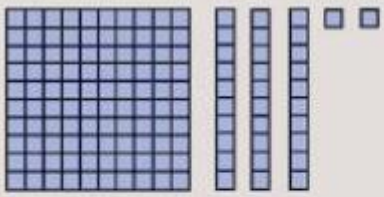
c



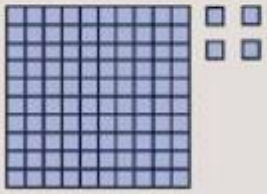
d



e



f



Exercise 1G

1 Which is larger?

a 0.4 or 0.39 **b** 0.06 or 0.1

c 1.2 or 1.13 **d** 1.02 or 1.3

e 2 or 1.64

2 Write down four numbers that are

a bigger than 3 but less than 4

b bigger than 3.4 but less than 3.6

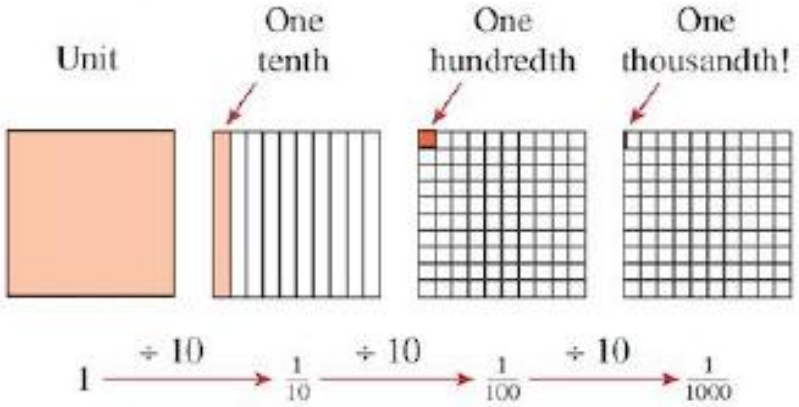
c bigger than 0.6 but less than 0.7

Thousandths

Each column heading is one tenth of the one to its left:

$$\text{Th} \xrightarrow{\div 10} \text{H} \xrightarrow{\div 10} \text{T} \xrightarrow{\div 10} \text{U}$$

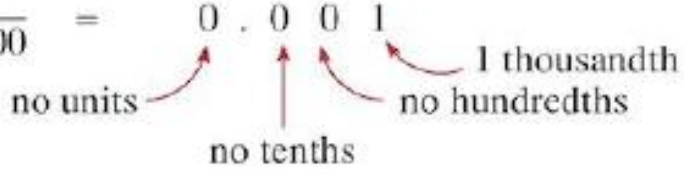
What happens if you divide the units column by 10?



The column to the right of $\frac{1}{100}$ column is $\frac{1}{1000}$ column.

Thousandths can be written as decimals:

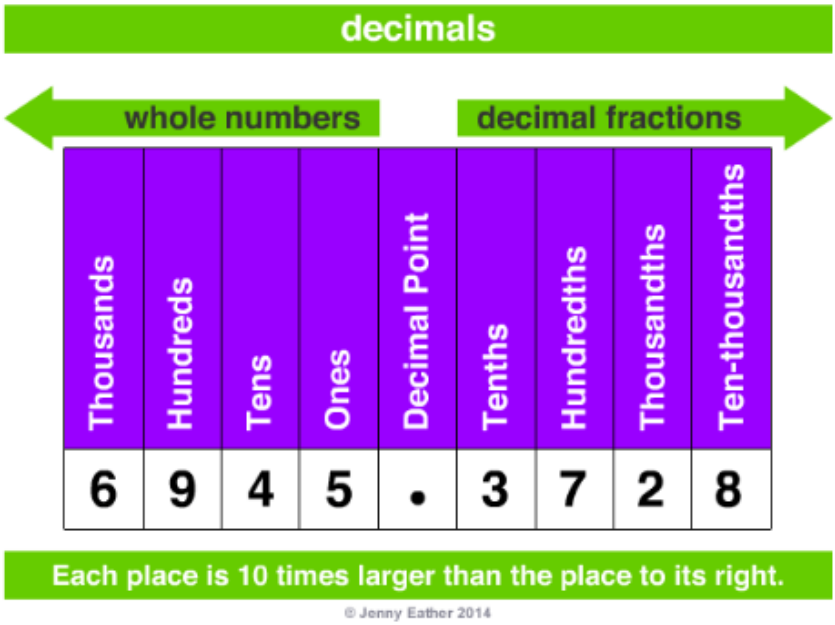
$$\frac{1}{1000} = 0.001$$



PLACE VALUE

- the value of a digit depending on its place in a number.
- in the decimal system, each place is 10x bigger than the place to its right.
- the decimal system uses 10 digits to show all numbers 0,1,2,3,4,5,6,7,8,9 by using place value and a decimal point to separate whole numbers from decimal fractions.

EXAMPLE:



place value number systems

- in a place value number system, powers of the base number are the basis of each place value column.

EXAMPLE:

base number

power, exponent, index, order

base

$b^4 = bxbxbxb$

place value number system

The number of units of the base number to the power of 5	The number of units of the base number to the power of 4	The number of units of the base number to the power of 3	The number of units of the base number to the power of 2	The number of units of the base number to the power of 1	The number of units of the base number to the power of 0
b^5	b^4	b^3	b^2	b^1	b^0

in the decimal system the base number is 10

10^5	10^4	10^3	10^2	10^1	10^0
100000's	10000's	1000's	100's	10's	1's
hundred thousands	ten thousands	thousands	hundreds	tens	ones

10^3	10^2	10^1	10^0	10^{-1}	10^{-2}	10^{-3}
1000	100	10	1	0.1	0.01	0.001

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Exercise 1I

1 Give the value of each underlined digit.

a 1̲3.3 **b** 6̲2.4 **c** 12.8̲4

d 0.03̲2 **e** 1.24̲5 **f** 13.4̲6

g 11.80̲4

2 Write these numbers in order of size, smallest first.

a 0.3, 0.02, 0.4, 0.006

b 3, 30, 0.3, 0.003

c 36, 3.6, 0.36, 0.036

3 Write down four numbers that lie between:

a 3.14 and 3.15

b 0.68 and 0.69

4 The batting averages of five international batsmen are shown in this table:

Kallis	54.66
Tendulkar	54.72
Gambhir	56.00
Hussey	53.04
Ponting	55.67

Put them in order, starting with the highest average.

5 Write these numbers in order of size, smallest first

a 12.5, 14, 6.35, 10.2, 8.323, 0.099

b 2.44, 2.5, 2.501, 2.41, 2.412, 2.4

c 0.4, 0.04, 0.44, 0.444, 0.404, 0.044

6 Write these numbers in order of size, largest first

a 1, 0.9, 1.19, 0.119, 1.119, 1.1

b 7.703, 7.7, 7.65, 7.651, 7.61, 7.73

c 12.1, 12.101, 12.03, 12.042, 12.01, 12.11

SCIENTIFIC NOTATION

Writing numbers in scientific notation.

1. Put a decimal point after the first digit.
2. To find what power of 10 to use, count the number of places from the decimal point to the end of the number.
3. Drop the zeros and write the number x 10 to the power.

$$3.456000000000^{12}$$

$$= 3.456 \times 10^{12}$$

For very small numbers, count back to the original decimal point and write the power of 10 as a negative.

$$0.^{8}0000000725$$

$$= 7.25 \times 10^{-8}$$

examples

decimal notation	scientific notation
84 500	8.45×10^{4}
678 345.96	6.7834596×10^{5}
50 000 000 000	5×10^{10}
0.0246	2.46×10^{-2}
0.000 024 6	2.46×10^{-5}
0.000 000 000 1	1×10^{-10}

Write 320 000 in standard form.

3.2	Start by finding the number between 1 and 10 that has the same digits in the same order as the original number. Here, the extra 4 zero digits can be excluded because they do not change the size of your new number.
5 4 3 2 1 3 2 0 0 0 0.0 3.2	Now compare the position of the first digit in both numbers: '3' has to move 5 place orders to the left to get from the new number to the original number.
$320\,000 = 3.2 \times 10^5$	The first digit, '3', has moved five places. So, you multiply by 10^5.

Solve and give your answer in standard form.

a $(3\times10^5)\times(2\times10^6)$ **b** $(2\times10^3)\times(8\times10^7)$

c $(2.8\times10^6)\div(1.4\times10^4)$ **d** $(9\times10^6)\div(3\times10^8)$

a

$$(3\times10^5)\times(2\times10^6) = (3\times2)\times(10^5\times10^6)$$
$$= 6\times10^{5+6}$$
$$= 6\times10^{11}$$

Simplify by putting like terms together. Use the laws of indices where appropriate.

Write the number in standard form.

You may be asked to convert your answer to an ordinary number. To convert 6×10^{11} into an ordinary number, the '6' needs to move 11 places to the left:

6.0×10^{11}

11 10 9 8 7 6 5 4 3 2 1

$= 6\ 0\ 0\ 0\ 0\ 0\ 0\ 0\ 0\ 0\ 0\ 0.0$

b

$$(2\times10^3)\times(8\times10^7) = (2\times8)\times(10^3\times10^7)$$
$$= 16\times10^{10}$$

$$16\times10^{10} = 1.6\times10\times10^{10}$$
$$= 1.6\times10^{11}$$

The answer 16×10^{10} is numerically correct but it is not in standard form because 16 is not between 1 and 10. You can change it to standard form by thinking of 16 as 1.6×10.

c

$$(2.8\times10^6)\div(1.4\times10^4) = \frac{2.8\times10^6}{1.4\times10^4} = \frac{2.8}{1.4}\times\frac{10^6}{10^4}$$
$$= 2\times10^{6-4}$$
$$= 2\times10^2$$

Simplify by putting like terms together. Use the laws of indices.

d $(9\times10^6)+(3\times10^8)$

When adding or subtracting numbers in standard form it is often easiest to re-write them both as ordinary numbers first, then convert the answer to standard form.

$9\times10^6 = 9000000$

$3\times10^8 = 300000000$

So $(9\times10^6)+(3\times10^8) = 300000000+9000000$

$= 309000000$

$= 3.09\times10^8$

1 Write each of the following numbers in standard form.

a 380 **b** 4200000 **c** 45600000000 **d** 65400000000000

e 20 **f** 10 **g** 10.3 **h** 5

2 Write each of the following as an ordinary number.

a 2.4×10^6 **b** 3.1×10^8 **c** 1.05×10^7 **d** 9.9×10^3 **e** 7.1×10^1

3 Simplify each of the following, leaving your answer in standard form.

a $(2\times10^{13})\times(4\times10^{17})$ **b** $(1.4\times10^8)\times(3\times10^4)$ **c** $(1.5\times10^{13})^2$

d $(12\times10^5)\times(11\times10^2)$ **e** $(0.2\times10^{17})\times(0.7\times10^{16})$ **f** $(9\times10^{17})\div(3\times10^{16})$

g $(8\times10^{17})\div(4\times10^{16})$ **h** $(1.5\times10^8)\div(5\times10^4)$ **i** $(2.4\times10^{64})\div(8\times10^{21})$

j $(1.44\times10^7)\div(1.2\times10^4)$ **k** $\dfrac{(1.7\times10^8)}{(3.4\times10^5)}$ **l** $(4.9\times10^5)\times(3.6\times10^9)$

4 Simplify each of the following, leaving your answer in standard form.

a $(3\times10^4)+(4\times10^3)$ **b** $(4\times10^6)-(3\times10^5)$ **c** $(2.7\times10^3)+(5.6\times10^5)$

d $(7.1\times10^9)-(4.3\times10^7)$ **e** $(5.8\times10^9)-(2.7\times10^3)$

Write each of the following in standard form.

a 0.004 **b** 0.00000034 **c** $(2\times10^{-3})\times(5\times10^{-7})$

a

$0.0\overset{1}{\curvearrowright}0\overset{2}{\curvearrowright}0\overset{3}{\curvearrowright}4$

4.0

$= 4\times10^{-3}$

Start with a number between 1 and 10, in this case 4.

Compare the position of the first digit: '4' needs to move 3 place orders to the right to get from the new number to the original number. In worked example 17 you saw that moving 5 places to the *left* meant multiplying by 10^5, so it follows that moving 3 places to the *right* means multiply by 10^{-3}.

Notice also that the first non-zero digit in 0.004 is in the 3rd place after the decimal point and that the power of 10 is –3.

Alternatively: you know that you need to divide by 10 three times, so you can change it to a fractional index and then a negative index.

$0.004 = 4\div10^3$

$= 4\times10^{\frac{1}{3}}$

$= 4\times10^{-3}$.

b

$0.00000034 = 3.4\div10^7$

$= 3.4\times10^{-7}$

$0.\overset{1}{\curvearrowright}0\overset{2}{\curvearrowright}0\overset{3}{\curvearrowright}0\overset{4}{\curvearrowright}0\overset{5}{\curvearrowright}0\overset{6}{\curvearrowright}0\overset{7}{\curvearrowright}3\,4 = 3.4\times10^{-7}$

Notice that the first non-zero digit in 0.00000034 is in the 7th place after the decimal point and that the power of 10 is –7.

c

$(2\times10^{-3})\times(3\times10^{-7})$

$= (2\times3)\times(10^{-3}\times10^{-7})$

$= 6\times10^{-3+-7}$

$= 6\times10^{-10}$

Simplify by gathering like terms together.

Use the laws of indices.

1 Write each of the following numbers in standard form.

a 0.004 **b** 0.00005 **c** 0.000032 **d** 0.0000000564

2 Write each of the following as an ordinary number.

a 3.6×10^{-4} **b** 1.6×10^{-8} **c** 2.03×10^{-7} **d** 8.8×10^{-3} **e** 7.1×10^{-1}

3 Simplify each of the following, leaving your answer in standard form.

a $(2\times10^{-4})\times(4\times10^{-16})$ **b** $(1.6\times10^{-8})\times(4\times10^{-4})$

c $(1.5\times10^{-6})\times(2.1\times10^{-3})$ **d** $(11\times10^{-5})\times(3\times10^{2})$

e $(9\times10^{17})\div(4.5\times10^{-16})$ **f** $(7\times10^{-21})\div(1\times10^{16})$

g $(4.5\times10^{8})\div(0.9\times10^{-4})$ **h** $(11\times10^{-5})\times(3\times10^{2})\div(2\times10^{-3})$

4 Simplify each of the following, leaving your answer in standard form.

a $(3.1\times10^{-4})+(2.7\times10^{-2})$ **b** $(3.2\times10^{-1})-(3.2\times10^{-2})$

c $(7.01\times10^{3})+(5.6\times10^{-1})$ **d** $(1.44\times10^{-5})-(2.33\times10^{-6})$

Applying your skills

5 Find the number of seconds in a day, giving your answer in standard form.

6 The speed of light is approximately 3×10^8 metres per second. How far will light travel in:

a 10 seconds **b** 20 seconds **c** 102 seconds

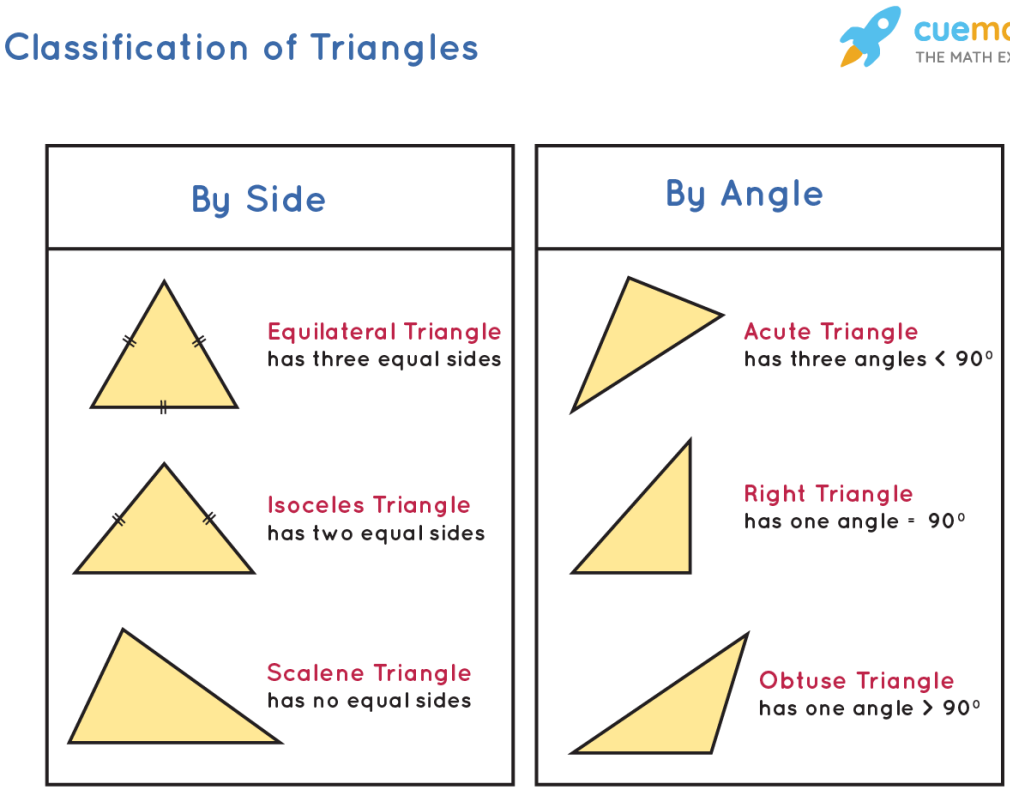
7 Data storage (in computers) is measured in gigabytes. One gigabyte is 2^{30} bytes.

a Write 2^{30} in standard form correct to 1 significant figure.

b There are 1024 gigabytes in a terabyte. How many bytes is this? Give your answer in standard form correct to one significant figure.

UNIT TWO: GEOMETRY AND MEASUREMENT

TRIANGLES

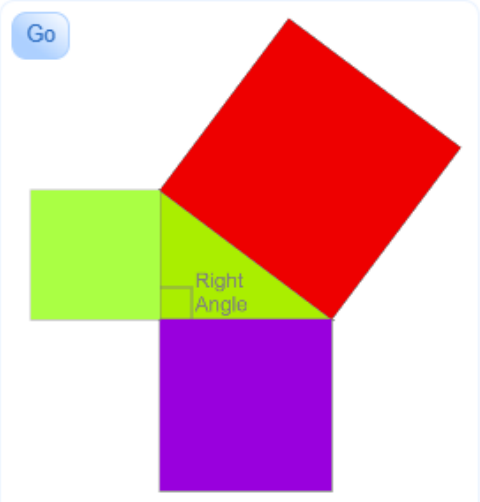


PYTHAGOREAS THEOREM

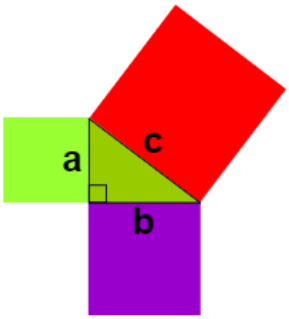
Watch the following video: http://shorturl.at/zCIV1

When a triangle has a right angle (90°) ...

... and squares are made on each of the three sides, ...

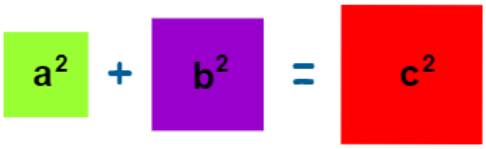


*... then the biggest square has the **exact same area** as the other two squares put together!*



It is called "Pythagoras' Theorem" and can be written in one short equation:

$$a^2 + b^2 = c^2$$

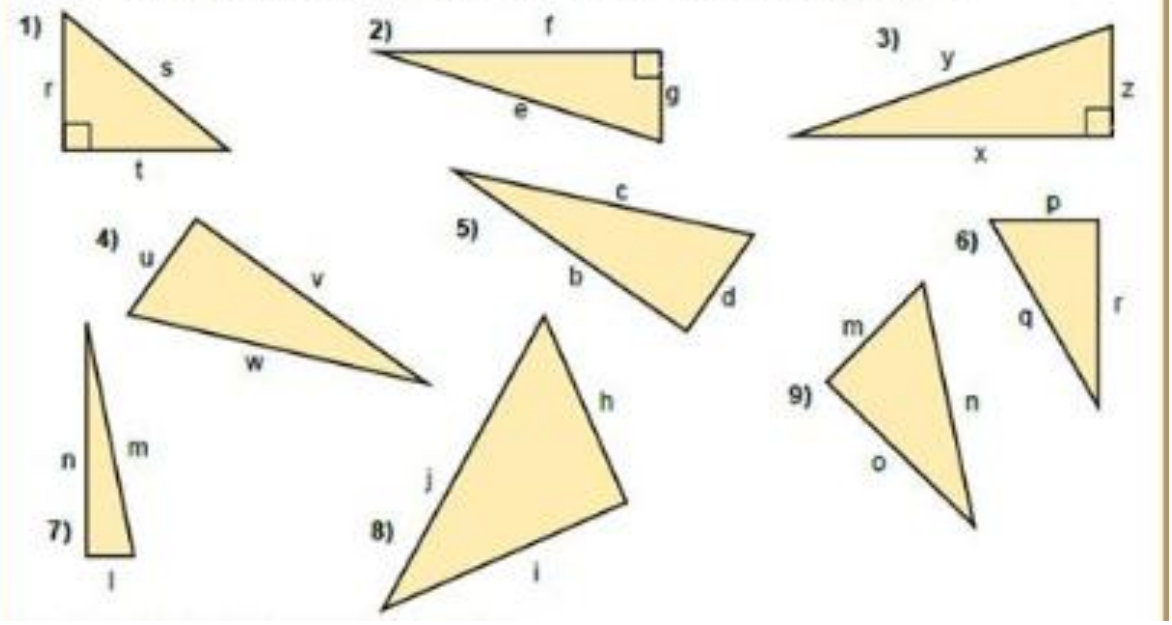


Definition

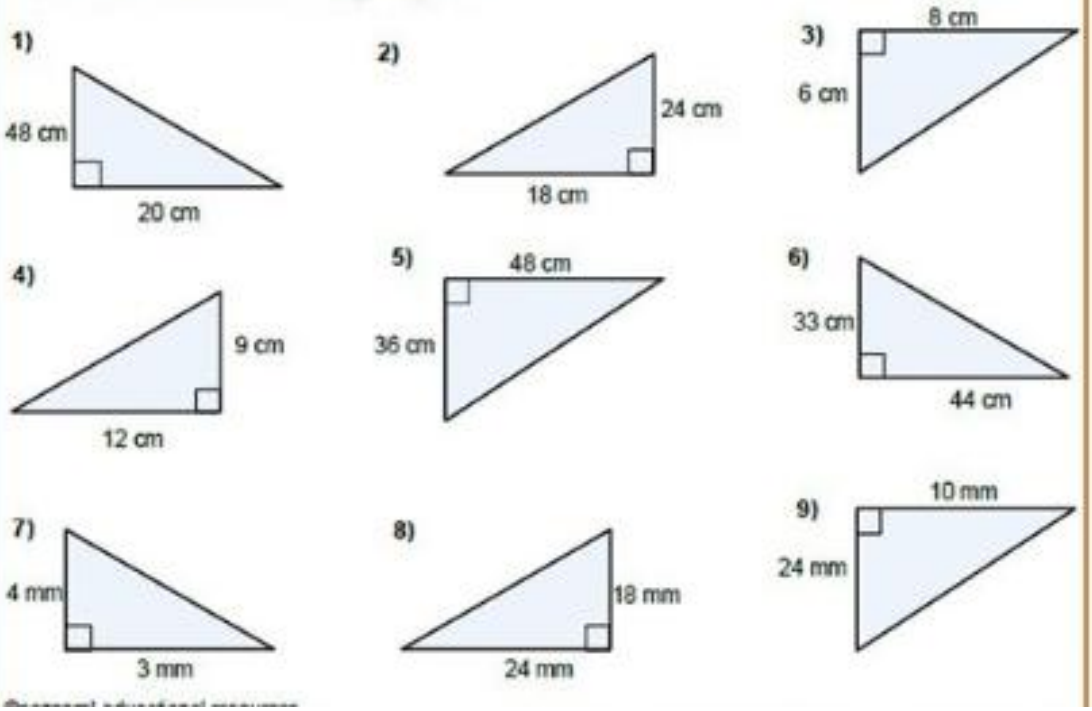
The longest side of the triangle is called the "hypotenuse", so the formal definition is:

In a right angled triangle:
the square of the hypotenuse is equal to
the sum of the squares of the other two sides.

Section A: *Put a circle around the letter which shows the hypotenuse.*



Section B: *Find the missing lengths.*



CONGRUENT OR SIMILAR TRIANGLES

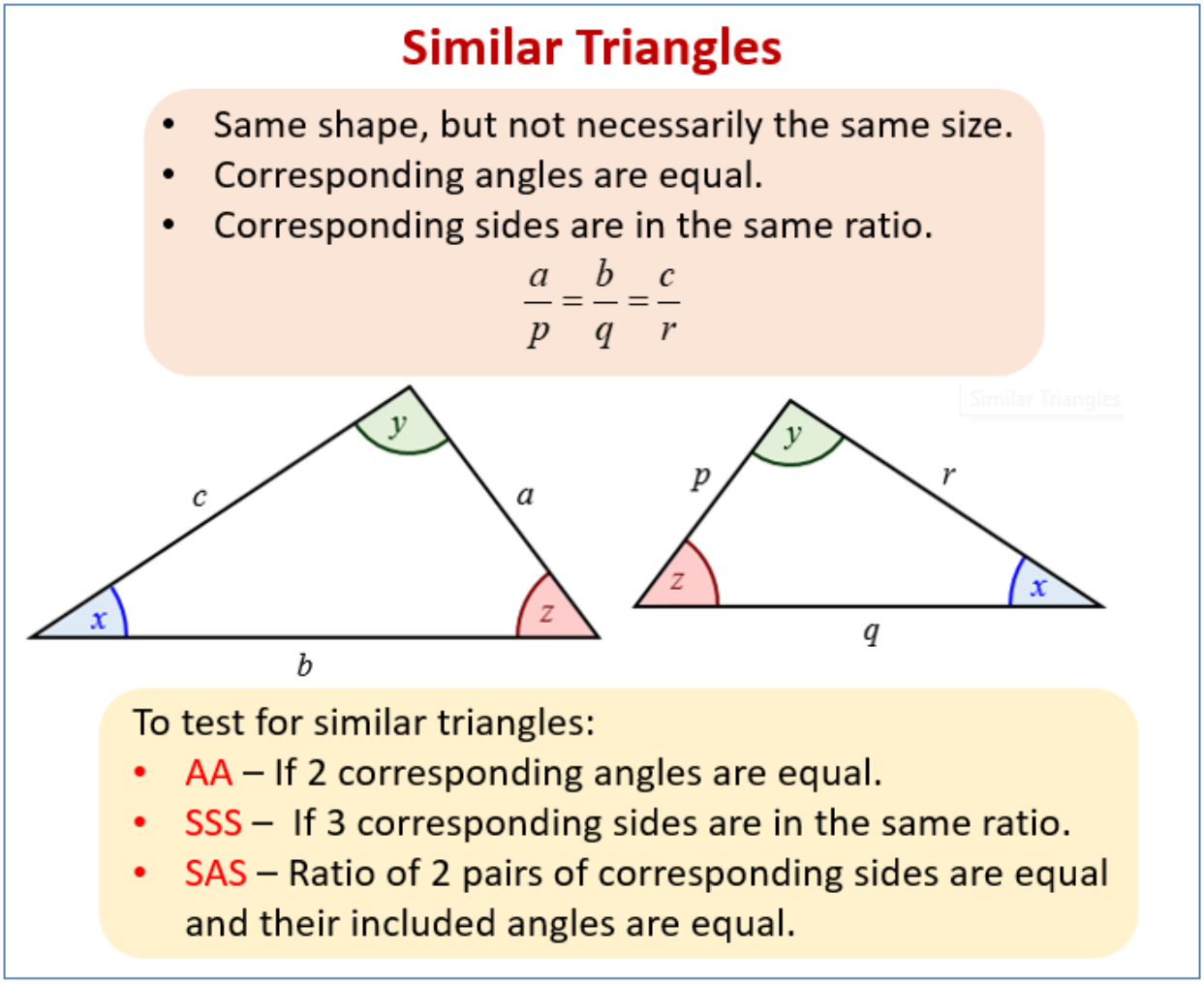
Geometry definition for Congruent:

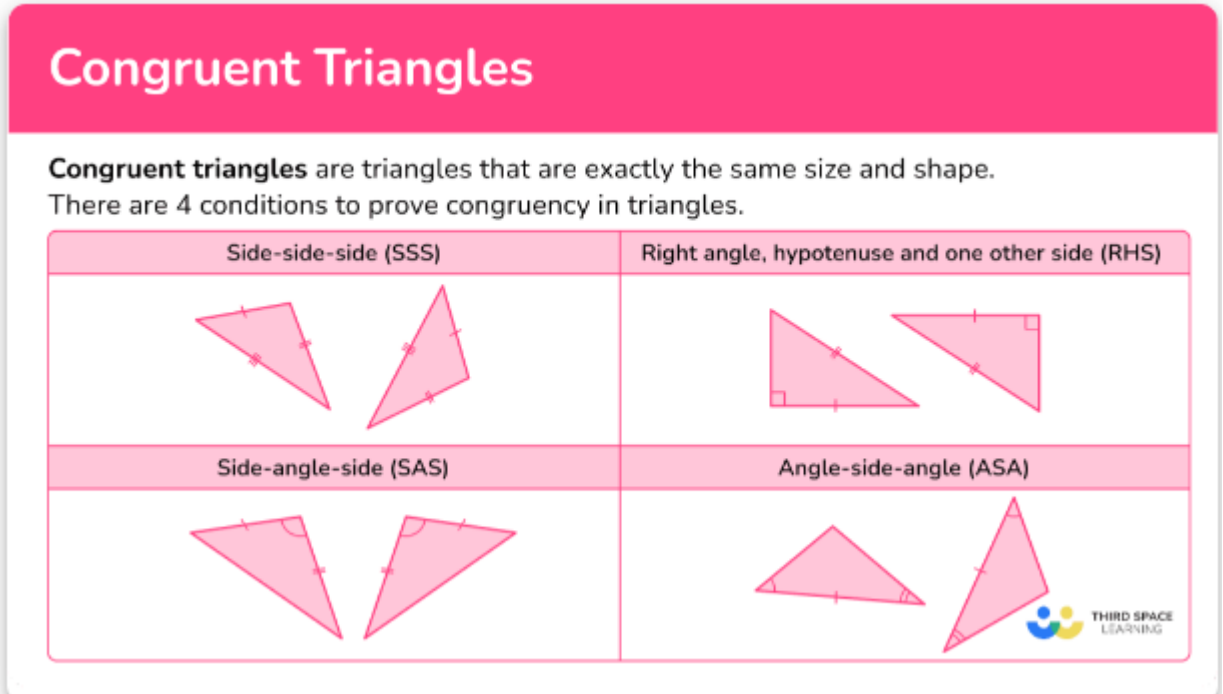
Two shapes are congruent if they have the same shape and the same size.

Geometry definition for Similar:

Two shapes are similar if they have the same shape, but not necessarily the same size.

The following diagrams gives the conditions for similar triangles.



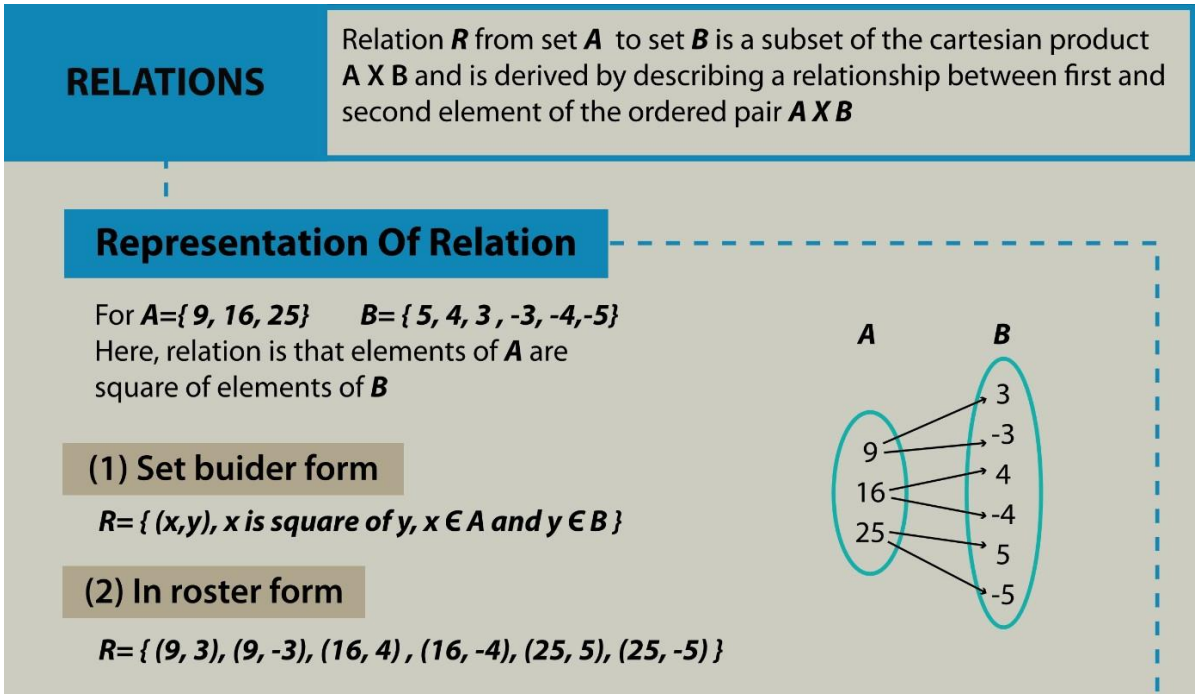


CONGRUENT TRIANGLES PROJECT:

https://www.youtube.com/watch?app=desktop&v=LRakFid_bXQ

UNIT 3: ALGEBRA AND FUNCTIONS

Given any two non-empty sets A and B, A relation R from A to B is a subset of the Cartesian product A x B and is derived by describing a relationship between the first element (say x) and the other element (say y) of the ordered pairs in A & B.



https://www.youtube.com/watch?v=52tpYl2tTqk&ab_channel=mathantics

A function is a **relation** between a set of inputs and a set of outputs in which each input is related with a **unique output**. A function is a rule that relates an input to exactly one output.

Functions

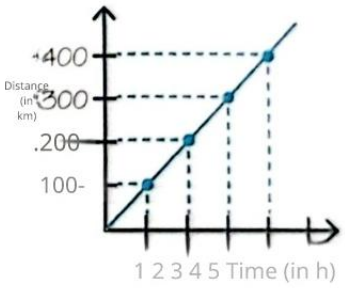
Function

A function (f) is defined as the relation between two sets A and B such that each value of the first is assigned a unique value of the second. These values are represented by ordered pairs (x;y) such that x E A and y € B.

· Domain (Dom f): is the set of values that the independent variable (x) can take.

· Range (Im f): is the set of values that the dependent variable (y) can take.

The graph shows the distance traveled as a function of time.



Independent variable: time (in hours). Dependent variable: distance traveled (in kilometers).

Graph analysis

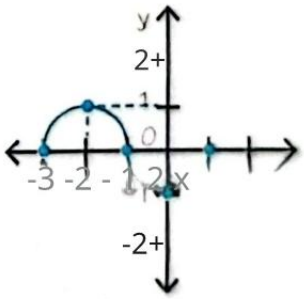
· Set of zeros (C°): are the values of x such that f(x) = 0. Graphically, they are the values at which the function intersects the x-axis.

· Set of positivity (C+): are the values for which f(x) > 0 and the graph is above the x-axis.

· Set of negativity (C-): are the values for which f(x) < 0 and the graph is below the x-axis.

· Interval of increase (11): is the set of values of x for which the function is increasing.

· Interval of decrease (14) is the set of values of x for which the function is decreasing.



Roots: x = - 3, x =- 1 and x = 1

C°= {-3; - 1;1}

C+ =(-3 ;- 1)

C =(-1;1)

IT=(-3;2) U (0:1)

IV = (-2;0)

Max. = (-2;1)

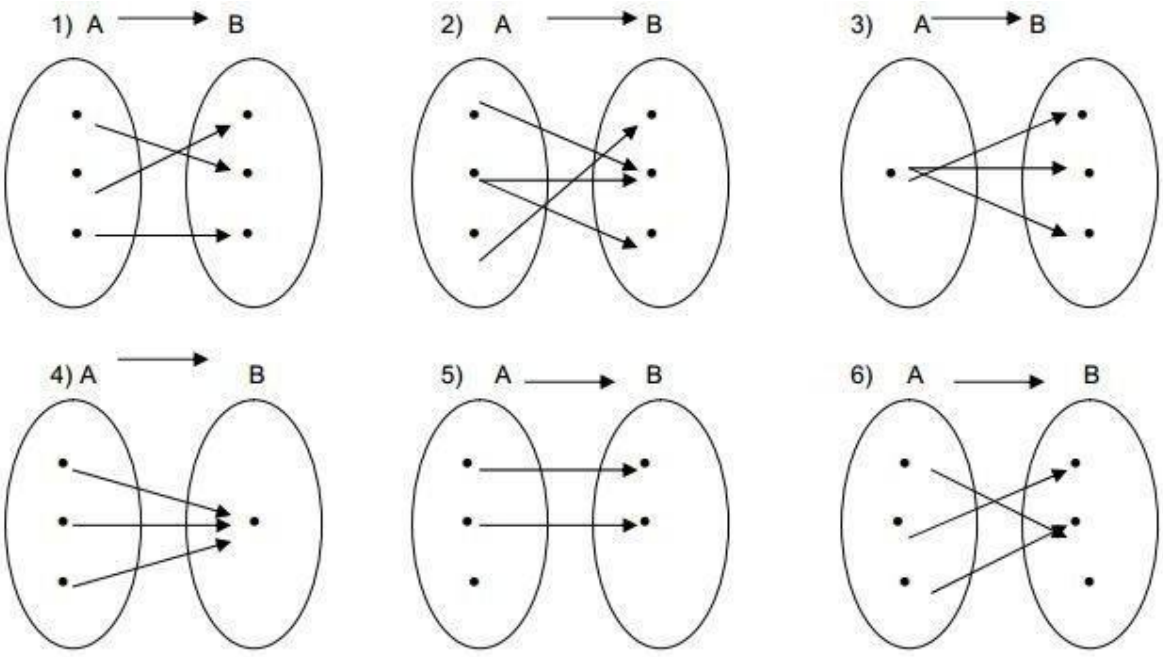
Min. = (0 ;- 1)

The relative extrema are those points where the function neither increases nor decreases:

· Maximum: is the highest point on the graph.

· Minimum: is the lowest point on the graph.

a) Exercise: Identify if the following relations are or are not functions. Justify your answer.



A function can be thought of as a rule or formula that tells how to associate the elements in one set (the ***domain***) with the elements in another set (the ***range***).

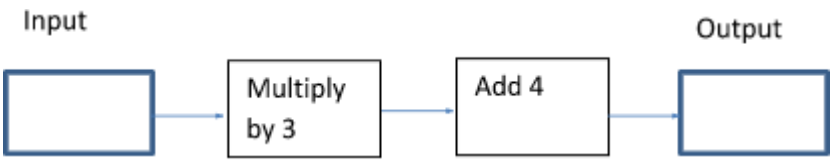
TERMINOLOGIES:

Before getting into details, let us get familiar with a few terms:

- **Image**: Suppose we are looking in a mirror. What do we see? An image or reflection. Similarly, for any ordered pairs, in any Cartesian product (say A × B), the second element is called the image of the first element.
- **Domain**: The set of all first elements of the ordered pairs in a relation R from a set A to a set B.
- **Range**: The set of all second elements in a relation R from a set A to a set B.

We should be familiar with function notation.

This is a function machine:



REVIEW

For example:

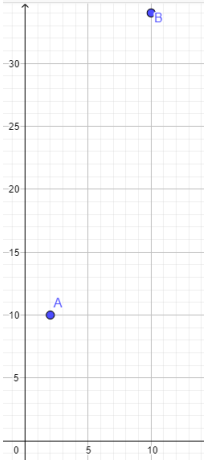
If 7 is input, then $7\times3+4=25$ is output.

A number enters the function machine is called the **input**. When a function machine uses a function (in this case *multiply by 3 then add 4*) it produces an **output**. Here, if the input is 2 then the output is 10 because $2\times3+4=10$. If the input is 10 the output is 34, and so on. We can use a mapping diagram like this one to show what outputs go with particular inputs.

Or it can be shown in a table:

Input	Output
2	10
10	34

If we graph the input and output as locations, we can see that the input is x and the output is y. then, we will have two locations: (2,10) and (10,34).



The set of numbers in the input: $2,10$ is what in Math is called the DOMAIN. The set of numbers in the output: $10,34$ is the RANGE.

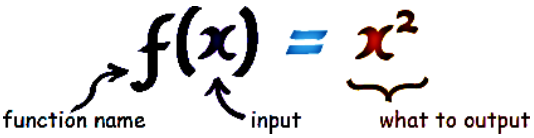


THEN, THE DOMAIN WILL BE THE SET OF **ALL THE VALUES** FOR ***X***

It is useful to give a function a **name**.

The most common name is "***f***", but we can have other names like "***g***" ... or even "***marmalade***" if we want.

But let's use "f":



We say *"f of x equals x squared"*

What goes **into** the function is put inside parentheses () after the name of the function:

So, ***f(x)*** shows us the function is called "***f***", and "***x***" goes **in**

And we usually see what a function does with the input:

f(x) = x² shows us that function "***f***" takes "***x***" and squares it.

Don't get too concerned about "x", it is just there to show us where the input goes and what happens to it. "x" is just a place holder, and can even be "x", "t", "q" or even "marmalade" too!

For example, we can think about a function that shows how many cakes will be baked during an amount of time: ct, where "c" calls for the amount of cupcakes and "t" will be how many minutes, hours, days, seconds… we took to bake.

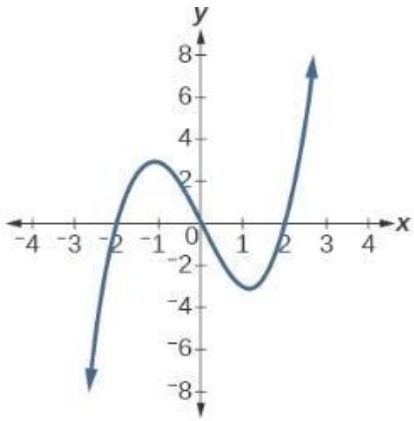
Now, decode the following functions:

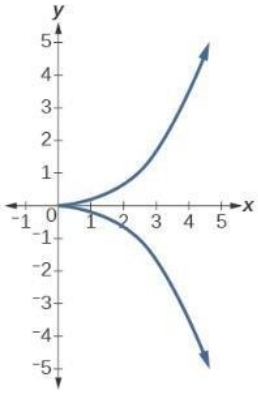
1. ft=2t-3 $f_{(7)}=$
2. fx=14-0.5x $f_{(30)}=$
3. ht=50-t5 $h_{(35)}=$
4. kx=6x+100 $k_{(-5)}=$
5. ht=-20+11t $h_{(11)}=$
6. gr=-5r+13 $g_{(3)}=$
7. fx=5x-3 $f_{(5)}=$
8. kt=13t-2 $k_{(3)}=$
9. hx=17+x6 $h_{(-18)}=$
10. gr=-1-7r $g_{(6)}=$

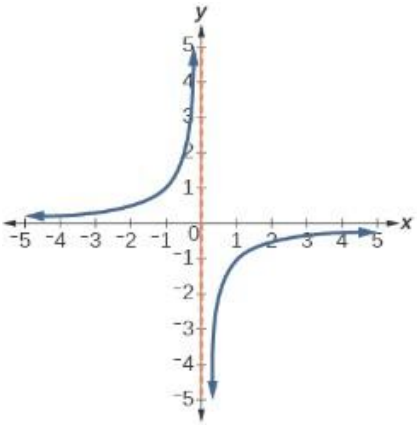
EXERCISES

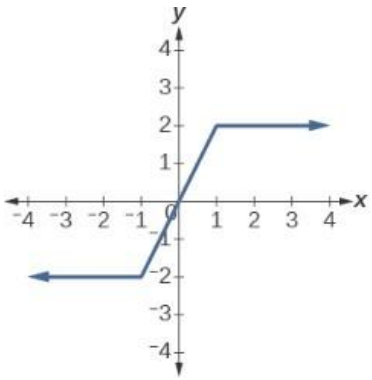
a) What is the difference between a relation and a function?
b) What is the difference between the input and the output of a function?
c) Why does the vertical line test tell us whether the graph of a relation represents a function?
d) Determine whether the relation represents a function.
{(a,b),(c,d),(a,c)}
e) Determine whether the relation represents a function.
{(a,b),(b,c),(c,c)}

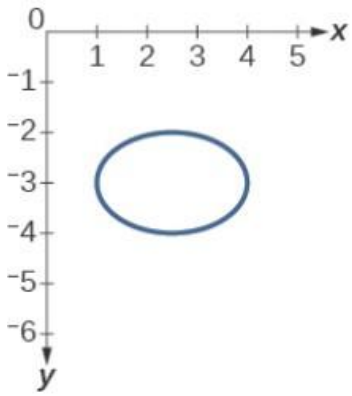
f) Use the vertical line test to determine which graphs show relations that are functions.

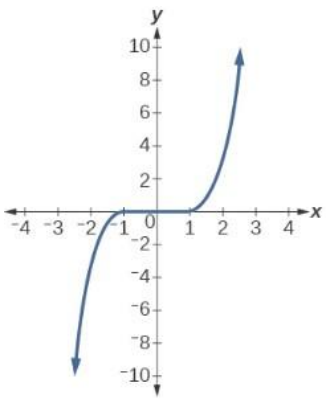




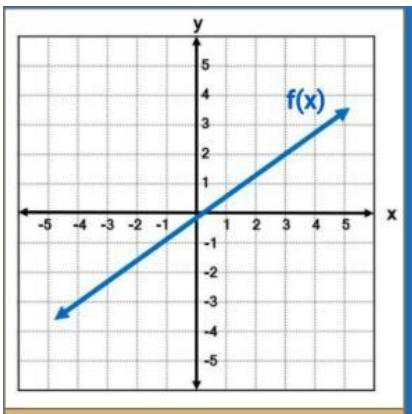






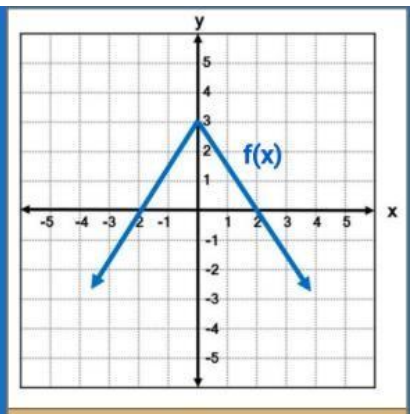


g) Evaluate each function



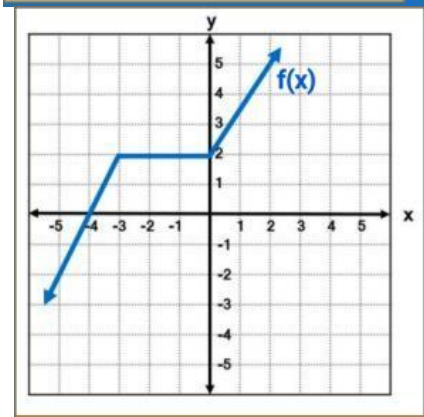
$f(3) =$ _____

$f(x) = -3,\ x =$ _____



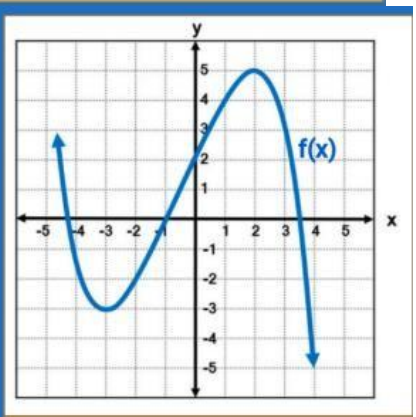
$f(2) =$ _____

$f(x) = 3,\ x =$ _____



$f(-1) =$ _____

$f(x) = -2,\ x =$ _____

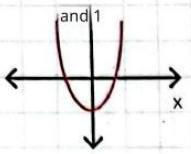


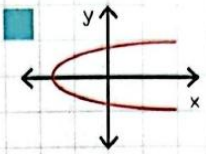
$f(0) =$ _____

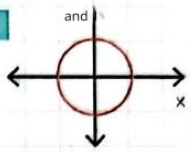
$f(x) = 5,\ x =$ _____

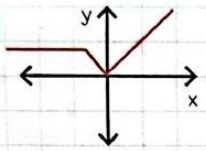
Function Analysis

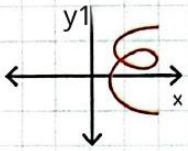
1. Mark with an X the graphs that correspond to functions.

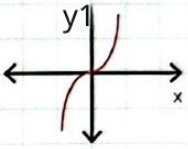
a. 

C. 

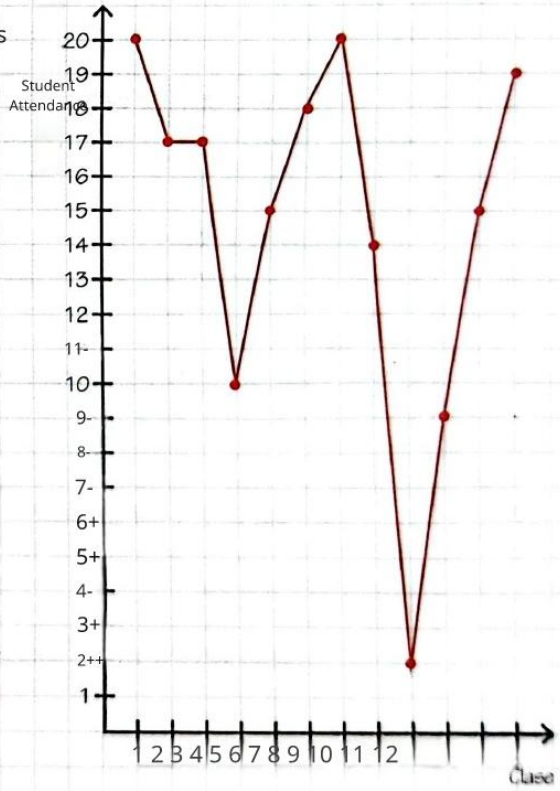
e. 

b. 

d. 

f. 

2. Observe the graph and answer. The graph shows the attendance of the 20 students enrolled in an English course during 12 classes.



a. What is the dependent variable?
And the independent one?

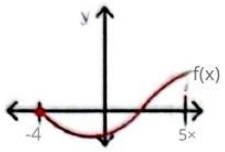
b. In which classes was there higher attendance?

c. Which class had the lowest attendance?

d. What is the average attendance?

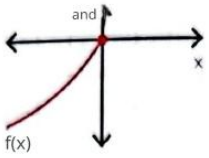
3 Indicate the domain of each function.

a.



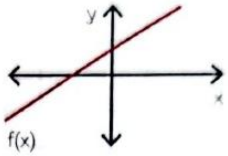
Dom f(x) = ...

b.



Dom f(x) = ...

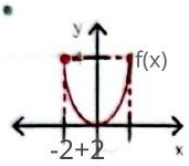
c.

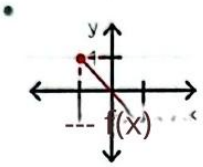


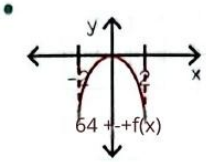
Dom f(x) = ...

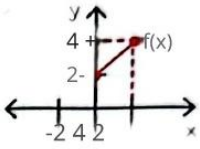
4 Connect each image set to its graph with an arrow.

a. Imf= [-1;1] b. Imf= [0;4] c. Imf= [2;4] d. Imf= [-4;0]









5 Observe the graph and answer. The graph shows the car sales of a dealership over a year. a. What is the independent variable? And the dependent one?



b. What is the domain of the function?

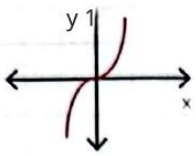
c. What is the image?

d. What was the quarter with the highest sales of the year?

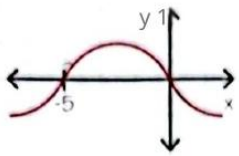
e. In which month were there no sales?

f. How many cars did the dealership sell up to and including the month of August?

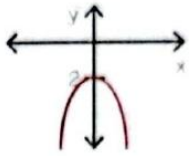
6 Mark the set of positivity (C+) of each function with an X. a. b. C.



C+= (-00;0)
C+= (0;+00)



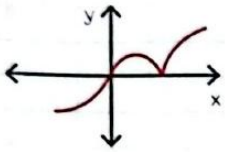
C+ = (-5;0)
C+ = (-5;+0)



C+=Ø C+ =
(-00;2)

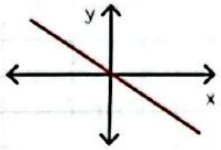
7 Indicate the set of negativity (C-) of each function, and then answer.

a.



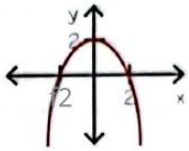
C -=

b.



C -=

c.



C -=

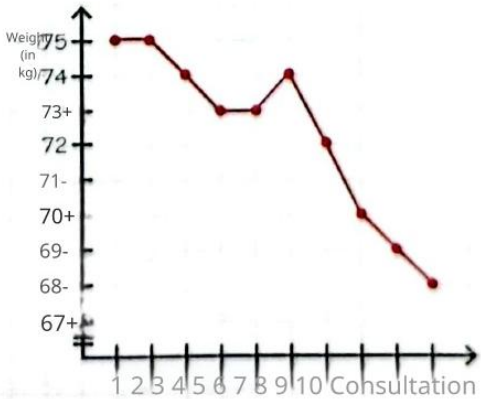
8 Observe the graph and solve.

The graph shows the weight variation of a person during a weight loss treatment.

a. What is the independent variable? And the dependent one?

b. What is the domain of the function?

c. What is the image?



d. Complete.

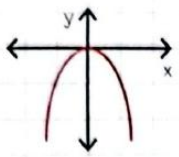
· C+ =

• C- =

e. What is the maximum of the function? And the minimum? What do they represent?

9 Complete the increasing (I) and decreasing (+) intervals for each function.

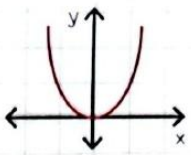
a.



I↑ =

I↓ =

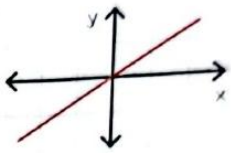
c.



I↑ =

I↓ =

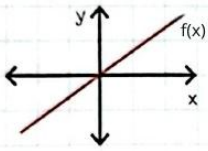
e.



I↑ =

I↓ =

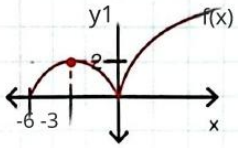
b.



I↑ =

I↓ =

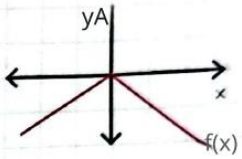
d.



I =

D=

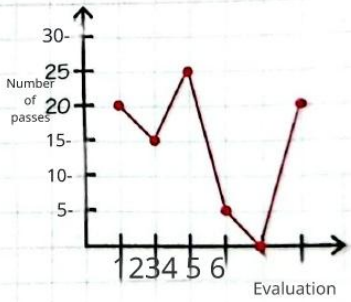
f.



I↑ =

I↓ =

10 Observe the graph and complete. Then answer. The graph shows the number of third-year students who passed the first six math evaluations.



a. Dom =

b. Im =

c. C+ =

d. C- =

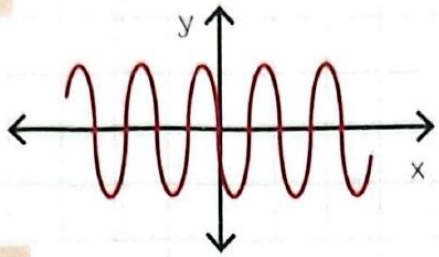
e. R =

f. I+ =

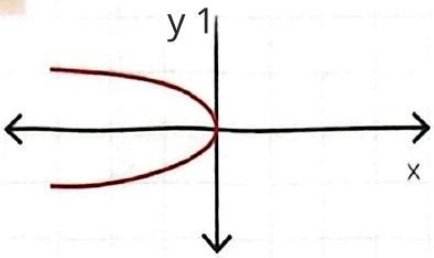
g. What was the maximum number of passes? And the minimum? In which evaluations?

1 Mark with an X the graphs that correspond to functions.

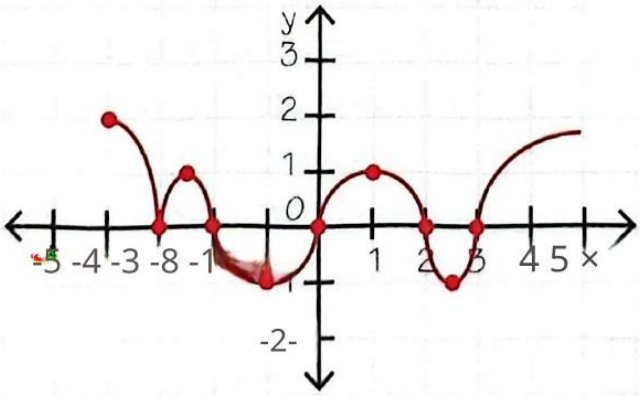
a.



b.



2 Observe the graph and complete.



a. Domain.

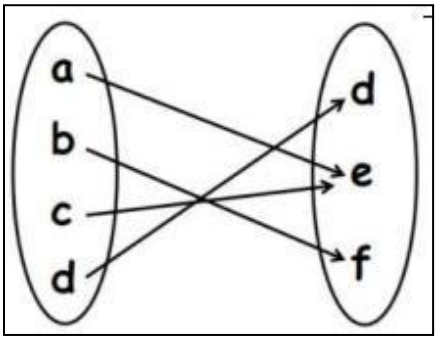
b. Range.

c. Sets of positivity and negativity.

d. Maxima and minima.

e. Intervals of increase and decrease.

2.1 LINEAR GRAPHS

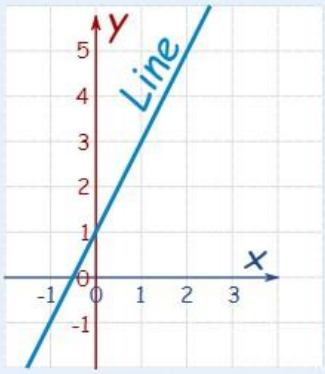


A set of points which can be joined by a straight line is called **linear graph**. The points on a straight line form a linear relationship and are generated by a linear function.

You can use the notation $f x=2x+1$ and $y=2x+1$ to refer to the same line, the first one is a *function*, the other, an *equation*.

Let us look more closely at one example:

Example: **y = 2x + 1** is a linear equation:



The graph of **y = 2x+1** is a straight line

- When x increases, y increases **twice as fast**, so we need 2x
- When x is 0, y is already 1. So +1 is also needed
- And so: y = 2x + 1

Different Forms

There are many ways of writing linear equations, but they usually have **constants** (like "2" or "c") and must have simple **variables** (like "x" or "y").

Examples: These are linear equations:

- ✓ $y = 3x - 6$
- ✓ $y - 2 = 3(x + 1)$
- ✓ $y + 2x - 2 = 0$
- ✓ $5x = 6$
- ✓ $y/2 = 3$

But the variables (like "x" or "y") in Linear Equations do **NOT** have:

- Exponents (like the 2 in x^2)
- Square roots, cube roots, etc

Examples: These are **NOT** linear equations:

- ✗ $y^2 - 2 = 0$
- ✗ $3\sqrt{x} - y = 6$
- ✗ $x^3/2 = 16$

2.2 Equation of a Straight Line

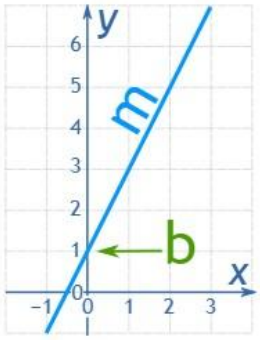
The equation of a straight line is usually written this way:

$$y = mx + b$$

m: Slope or Gradient

b: **y** value when **x=0** *(see Y Intercept)*

What does it stand for?

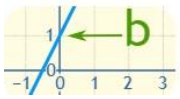


y = how far up

x = how far along

m = Slope or Gradient (how steep the line is)

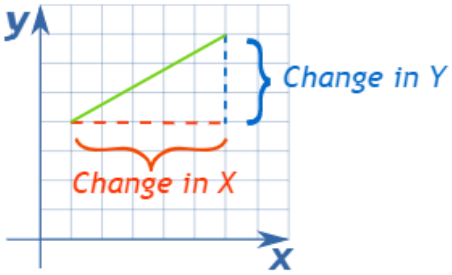
b = value of **y** when **x=0**



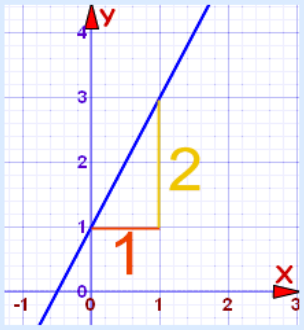
How do you find "a" and "b"?

- b is easy: just see where the line crosses the Y axis.
- a (the Slope) needs some calculation:

$$m = \frac{y_f - y_i}{x_f - } = \frac{\text{Change in Y}}{\text{Change in X}}$$



Example 1



$$m = \frac{2}{1} = 2$$

b = 1 *(value of* ***y*** *when x=0)*

Putting that into **y** = **mx** + **b** gets us:

$$y = 2x + 1$$

Positive or Negative Slope?

Going from left-to-right, the cyclist has to **P**ush on a **P**ositive Slope:

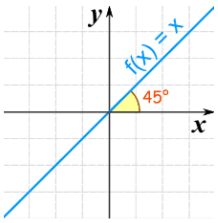


The Identity Function

There is a special linear function called the "Identity Function":

$$f(x) = x$$

And here is its graph:



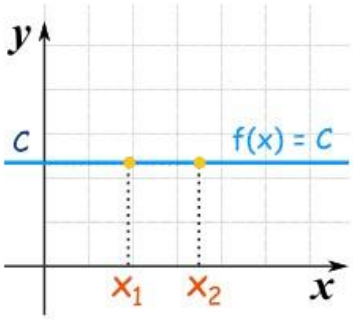
It makes a 45° (its slope is 1)

It is called "Identity" because what comes out is **identical** to what goes in:

In	Out
0	0
5	5
−2	−2
...etc	...etc

Constant Functions

Another special type of linear function is the Constant Function ... it is a horizontal line:

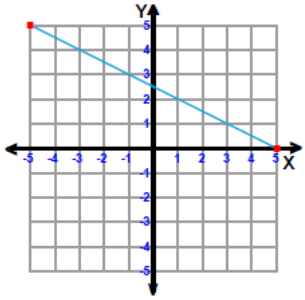


$$f(x) = C$$

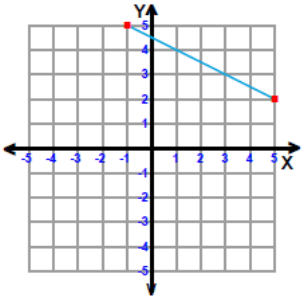
No matter what value of "x", f(x) is always equal to some constant value.

EXERCISES:

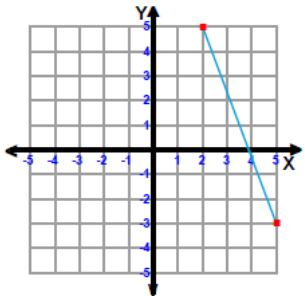
What is the slope of each line ?

1) 

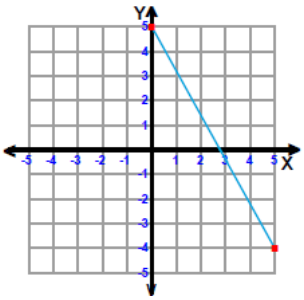
Slope = _____

2) 

Slope = _____

3) 

Slope = _____

4) 

Slope = _____

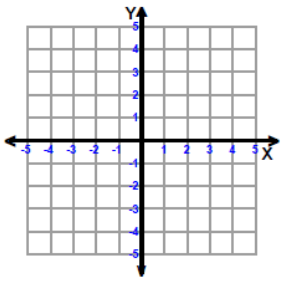
5) $y = \frac{5}{8}x + 3$ Slope = _____

6) $y = -5x + 2$ Slope = _____

7) $y = \frac{1}{2}x - 2$ Slope = _____

8) $y = -\frac{1}{8}x - 3$ Slope = _____

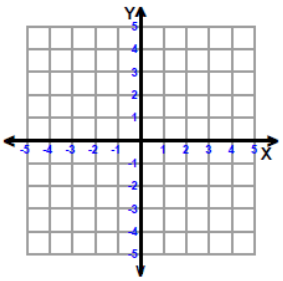
Write the slope-intercept form and plot the equation of each line given the slope and y-intercept.

9) 

Slope = $-\frac{1}{2}$

y-intercept = 2

Equation : _____

10) 

Slope = $-\frac{1}{3}$

y-intercept = 1

Equation : _____

Find the Slope from the Pair of Points

1) (1,-5) (3,5) slope = _____

2) (3,-2) (5,-4) slope = _____

3) (-5,-3) (5,0) slope = _____

4) (-4,-4) (-3,-2) slope = _____

5) (-1,-5) (-5,5) slope = _____

6) (-5,2) (5,1) slope = _____

7) (-5,-4) (5,-1) slope = _____

8) (-3,-2) (-2,2) slope = _____

9) (4,-4) (3,0) slope = _____

10) (2,-5) (-2,5) slope = _____

Find the Slope and Y-intercept for Each Equation

1) -6x + 5y = -10 slope = _____ y-intercept = _____

2) $y = -\frac{2}{5}x - 2$ slope = _____ y-intercept = _____

3) 4x + 7y = -14 slope = _____ y-intercept = _____

4) $y = \frac{7}{4}x - 3$ slope = _____ y-intercept = _____

5) -x +3y -6 slope = _____

y-intercept = _____

6) -2x + y = -4 slope = _____

y-intercept = _____

7) -5x + 2y = 6 slope = _____

y-intercept = _____

8) y = 3x + 2 slope = _____

y-intercept = _____

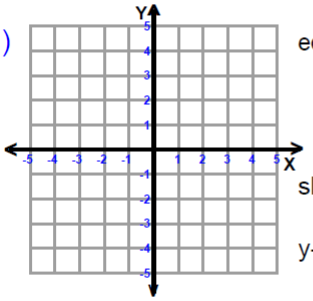
9) $y = \frac{1}{2}x + 4$ slope = _____

y-intercept = _____

10) y = - 6x - 3 slope = _____

y-intercept = _____

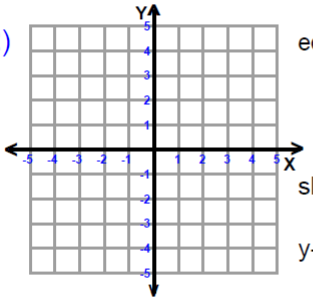
Sketch Each Line and Find the Slope and Y-intercept

1) 

equation $y = -\frac{8}{3}x + 4$

slope (m) = _____

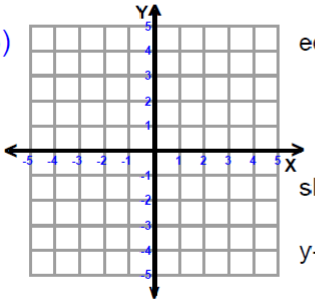
y-intercept = _____

2) 

equation $y = -\frac{1}{3}x + 1$

slope (m) = _____

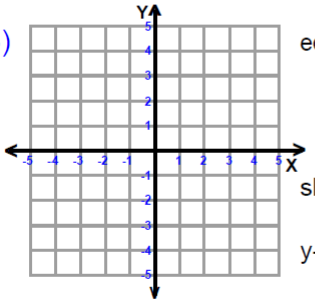
y-intercept = _____

3) 

equation $y = -\frac{7}{6}x + 2$

slope (m) = _____

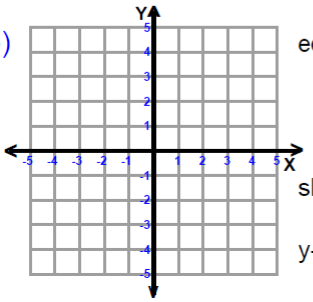
y-intercept = _____

4) 

equation $y = \frac{1}{4}x - 2$

slope (m) = _____

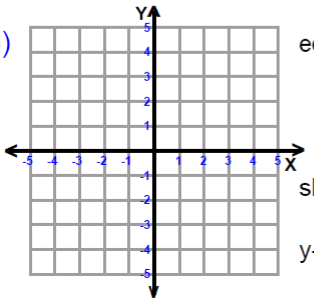
y-intercept = _____

5) 

equation y = 2x - 2

slope (m) = _____

y-intercept = _____

6) 

equation y = - x + 4

slope (m) = _____

y-intercept = _____

UNIT FOUR: DATA ANALYSIS, PROBABILITY AND STADISTICS

MEAN, MEDIAN, AND MODE

Watch the following video:

http://shorturl.at/gkpF2

MEAN OR AVERAGE

Mean: Add up the numbers and divide by how many numbers.

Example: what is the central value of 3, 7 and 8?

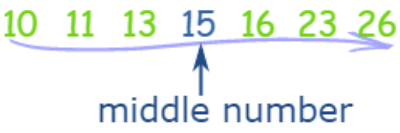
Answer: We calculate it by adding 3, 7 and 8 and then dividing the results by 3 (because there are 3 numbers):

(3+7+8) / 3 = 18/3 = 6



MEDIAN VALUE

The Median is the ***"middle"*** of a sorted list of numbers.



How to Find the Median Value

To find the Median, place the numbers in **value order** and find the **middle**.

Example: find the Median of **12, 3 and 5**

Put them in order:

3, 5, 12

The middle is **5**, so the median is **5**.

Two Numbers in the Middle

BUT, with an **even amount of numbers** things are slightly different.

In that case we find the **middle pair** of numbers, and then find the value that is **half way** between them. This is easily done by adding them together and dividing by two.

Mode

The mode is simply the number which appears **most often**.

Example: 6, 3, 9, 6, 6, 5, 9, 3

In {6, 3, 9, 6, 6, 5, 9, 3} the Mode is 6, as it occurs most often.



More Than One Mode

We can have more than one mode.

Example: {1, 3, 3, 3, 4, 4, 6, 6, 6, 9}

3 appears three times, as does 6.

So there are two modes: at **3** and **6**

Having two modes is called **"bimodal"**.

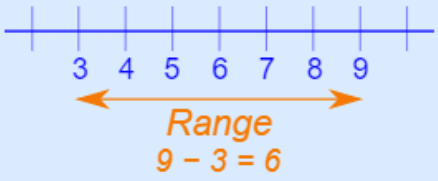
Having more than two modes is called **"multimodal"**.

RANGE

The Range is the difference between the lowest and highest values.

Example: In **{4, 6, 9, 3, 7}** the lowest value is 3, and the highest is 9.

So the range is 9 − 3 = **6**.



Find the mean, median, mode and range in each of the sets of data.
The first one has been done for you.

1)	15, 23, 19, 20, 23		5)	22, 37, 19, 25, 37, 51, 82	
order	15, 19, 20, 23, 23		order		
	Mean 100÷5=<u>20</u>	Median <u>20</u>		Mean	Median
	Mode <u>23</u>	Range 23-15=<u>8</u>		Mode	Range
2)	2, 7, 4, 2, 3, 6, 11		6)	6, 2, 13, 7, 6, 11, 10, 6, 2	
order			order		
	Mean	Median		Mean	Median
	Mode	Range		Mode	Range

3)	70, 63, 67, 62, 63		7)	109, 104, 96, 103, 104, 107, 98	
order			order		
	Mean	Median		Mean	Median
	Mode	Range		Mode	Range
4)	11, 4, 7, 8, 2, 6, 4		8)	14, 68, 38, 65, 36, 57, 65	
order			order		
	Mean	Median		Mean	Median
	Mode	Range		Mode	Range

STADISTICAL GRAPHS

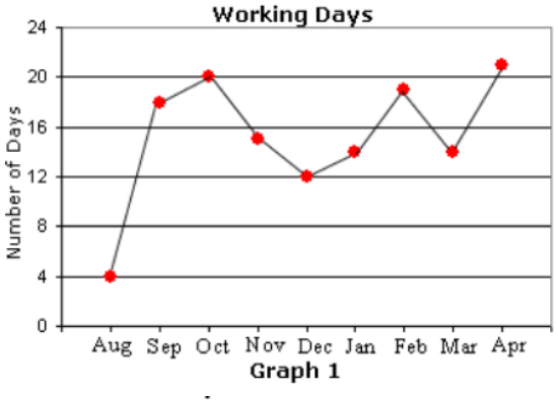
Watch the following video:

http://shorturl.at/loWZ9

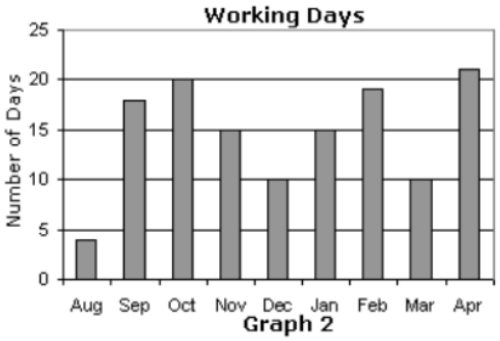
1. The table shows the number of school days each month during the last school year. Which graph best represents this data?

   a. Graph 1
   b. Graph 2
   c. Graph 3
   d. Graph 4

Month	Aug	Sep	Oct	Nov	Dec	Jan	Feb	Mar	Apr
Number of Days	4	18	20	15	12	14	19	14	21



Graph 1



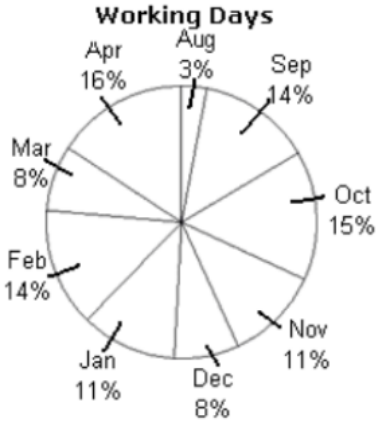
Graph 2

Working Days

Month	Number of School Days
August	●
September	● ● ● ●
October	● ● ● ● ●
November	● ● ● ●
December	● ● ●
January	● ● ● ●
February	● ● ● ● ●
March	● ● ●
April	● ● ● ● ●

● = 4 days

Graph 3



Graph 4

2. Identify the graph that is most appropriate to represent the amount of money spent on military over a period of time.
   a. Line graph
   b. Bar graph
   c. Pie graph
   d. Histogram
   e. Scatter plot

3. Which graph would be most appropriate to show the time in number of minutes that 25 employees take to travel to the office?
   a. Line graph
   b. Bar graph
   c. Pie graph
   d. Histogram
   e. Scatter plot

4. A survey was conducted on the length of the names of the students in a class. What type of graph would be most appropriate to show the range of the data?
   a. Line graph
   b. Bar graph
   c. Pie graph
   d. Histogram
   e. Scatter plot

5. The table shows the kinds of pets the students have in a class. What type of graph would be most appropriate for the data?

Pet	Frequency of the Pet
Birds	18
Cats	5
Dogs	23
Fish	17
Rabbits	6
Horse	4

   a. Line graph
   b. Bar graph
   c. Pie graph
   d. Histogram
   e. Scatter plot

6. The table shows the yearly average earnings (in thousands of dollars) for men and women. Which type of graph would be most appropriate for the data?

Year	1998	1999	2000	2001
Men	28.9	29.3	29.6	30.2
Women	19.5	20.4	22.3	22.4

   a. Line graph
   b. Bar graph
   c. Pie graph
   d. Histogram
   e. Scatter plot

7. The weight of 15 people(in kg) are listed: 34, 49, 23, 54, 35, 64, 40, 53, 52, 76, 48, 21, 47, 34, and 33. Which type of graph is appropriate for the data?

   a. Line graph
   b. Bar graph
   c. Pie graph
   d. Histogram
   e. Scatter plot

8. The table shows the number of cars sold in a week. Which type of graph is appropriate for the data?

   a. Line graph
   b. Bar graph
   c. Pie graph
   d. Histogram
   e. Scatter plot

Color of car	Red	White	Orange	Blue
Number of cars	15	24	7	11

9. The table shows the school enrollment between 1992 to 1996. Which type of graph is appropriate for the data?

Year	Number of students
1992	65
1993	85
1994	70
1995	70
1996	75

   a. Line graph
   b. Bar graph
   c. Pie graph
   d. Histogram
   e. Scatter plot

Combinations and Permutations

What's the Difference?

In English we use the word "combination" loosely, without thinking if the **order** of things is important.

So, in Mathematics we use more *precise* language:

- When the order doesn't matter, it is a **Combination**.
- When the order **does** matter it is a **Permutation**.



So, we should really call this a "Permutation Lock"!

In other words:

A Permutation is an **ordered** Combination.



To help you to remember, think "**P**ermutation ... **P**osition"

Permutations

There are basically two types of permutation:

- **Repetition is Allowed**: such as the lock above. It could be "333".
- **No Repetition**: for example the first three people in a running race. You can't be first *and* second.

1. Permutations with Repetition

These are the easiest to calculate.

When a thing has ***n*** different types ... we have ***n*** choices each time!

For example: choosing ***3*** of those things, the permutations are:

n × n × n
(n multiplied 3 times)

More generally: choosing ***r*** of something that has ***n*** different types, the permutations are:

n × n × ... *(r times)*

(In other words, there are **n** possibilities for the first choice, THEN there are **n** possibilites for the second choice, and so on, multplying each time.)

Which is easier to write down using an exponent of **r**:

n × n × ... (r times) = n^r

So, the formula is simply:

n^r
where ***n*** is the number of things to choose from, and we choose ***r*** of them, repetition is allowed, and order matters.

2. Permutations without Repetition

In this case, we have to **reduce** the number of available choices each time.

Without repetition our choices get reduced each time.

But how do we write that mathematically? Answer: we use the "factorial function"

The **factorial function** (symbol: !) just means to multiply a series of descending natural numbers. Examples:

- 4! = 4 × 3 × 2 × 1 = 24
- 7! = 7 × 6 × 5 × 4 × 3 × 2 × 1 = 5.040
- 1! = 1

Note: it is generally agreed that **0! = 1**. It may seem funny that multiplying no numbers together gets us 1, but it helps simplify a lot of equations.

The formula is written:

$$\frac{n!}{(n-r)!}$$

where ***n*** is the number of things to choose from,
and we choose ***r*** of them,
no repetitions,
order matters.

Notation

Instead of writing the whole formula, people use different notations such as these:

$$P(n,r) = {}^{n}P_{r} = {}_{n}P_{r} = \frac{n!}{(n-r)!}$$

Combinations

There are also two types of combinations (remember the order does **not** matter now):

- **Repetition is Allowed**: such as coins in your pocket (5,5,5.10,10)
- **No Repetition**: such as lottery numbers (2.14,15.27,30.33)

1. Combinations with Repetition

Actually, these are the hardest to explain, so we will come back to this later.

2. Combinations without Repetition

This is how lotteries work. The numbers are drawn one at a time, and if we have the lucky numbers (no matter what order) we win!

The easiest way to explain it is to:

- assume that the order does matter (ie permutations),
- then alter it so the order does **not** matter.

Notation

All these notations mean "n choose r":

$$C(n,r) = {}^{n}C_r = {}_{n}C_r = \binom{n}{r} = \frac{n!}{r!(n-r)!}$$

Just remember the formula:

$$\frac{n!}{r!(n-r)!}$$

In Conclusion

	Repeats allowed	No Repeats
Permutations (order matters):	n^r	$\frac{n!}{(n-r)!}$
Combinations (order doesn't matter):	$\frac{(r+n-1)!}{r!(n-1)!}$	$\frac{n!}{r!(n-r)!}$

Instructions for the Sports Uniform Design Project:

Project Objective: Design three sets of uniforms for a fictional sports team (hockey or soccer) that include specific variations for training, practice, and official matches. Apply concepts of permutations and combinations to create unique and distinctive options.

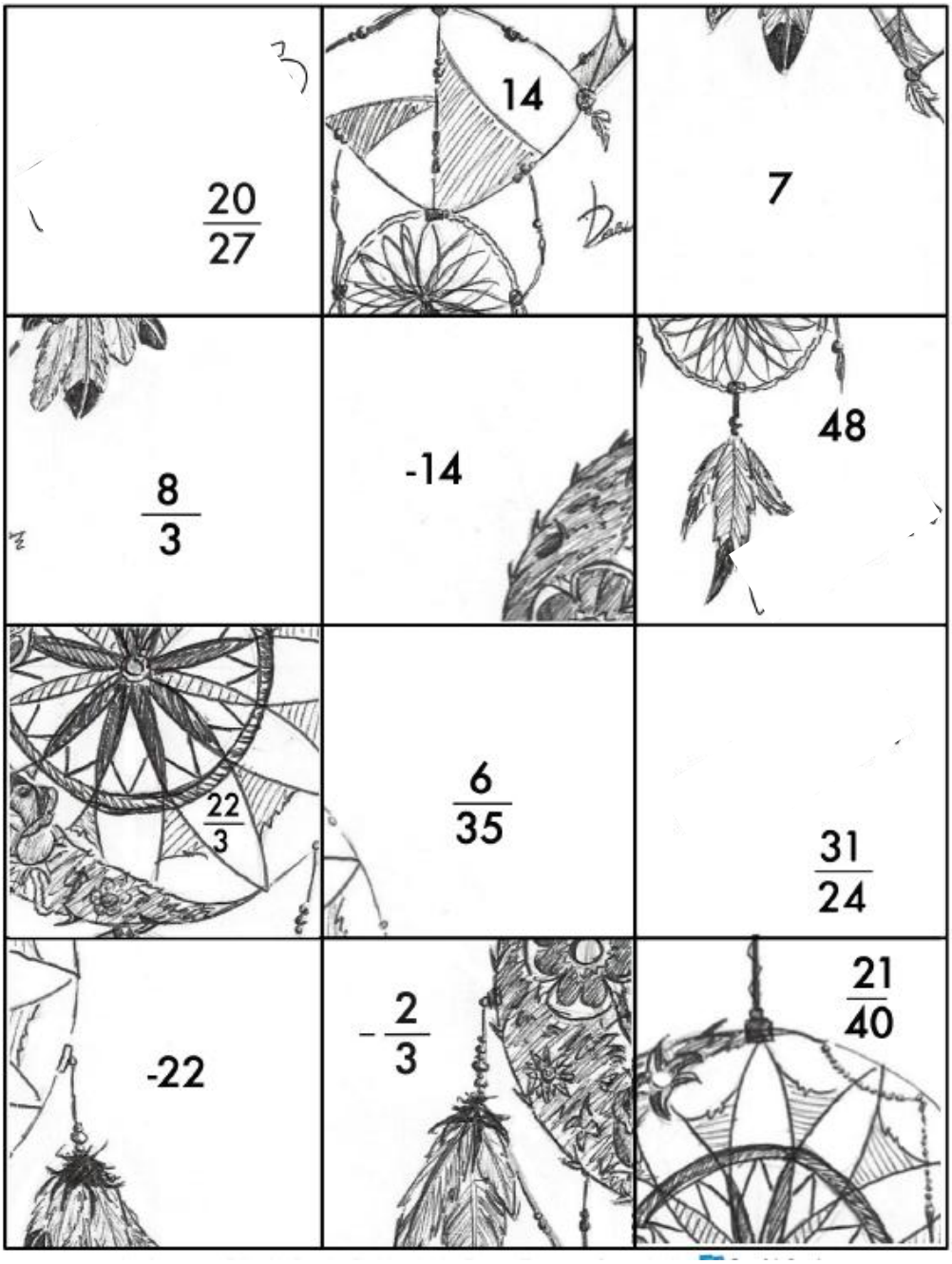
Steps:

1. **Team Selection:**
    - Choose a fictional hockey or soccer team for which you will design the uniforms. Consider colors, mascots, or distinctive elements that represent the team's identity.
2. **Training Uniform Design (15 minutes):**
    - Consider variations in colors and elements for the training uniform.
    - Determine how many unique combinations you can create for jerseys, pants, socks, and logos.
3. **Practice Uniform Design (15 minutes):**
    - Design the practice uniform with different combinations of colors and elements compared to the training uniform.
    - Ensure clear variation between the practice and training uniforms.
4. **Official Match Uniform Design (15 minutes):**
    - Focus creativity on designing the official match uniform.
    - Experiment with colors and elements to create a strong and representative identity for the team.
5. **Calculations and Justification (10 minutes):**
    - Perform calculations to determine how many unique permutations and combinations you have created in total for the three uniform sets.

- Justify design decisions by explaining how you applied permutations and combinations concepts in each set.

6. **Presentation (10 minutes):**
    - Prepare a brief presentation to share your designs with the class.
    - Highlight key variations and explain the reasoning behind each uniform set.
7. **Discussion and Feedback (15 minutes):**
    - After each presentation, engage in a group discussion about the strategies used and the application of mathematical concepts.
    - Provide constructive feedback on creativity and coherence in the designs.
8. **Final Reflection (5 minutes):**
    - Reflect on how the application of permutations and combinations influenced the creation of sports uniforms and the team's identity.

ACTIVITIES


$\frac{20}{27}$
14
7
$\frac{8}{3}$
-14
48
$\frac{22}{3}$
$\frac{6}{35}$
$\frac{31}{24}$
-22
$-\frac{2}{3}$
$\frac{21}{40}$